First principles evidence of stiffening during spinodal decomposition in FeCr

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Abstract

We explain stiffening during spinodal decomposition in bcc δ-ferrite previously observed in cast austenitic stainless steels by combining experimental measurements with first-principles simulations of bcc $Fe_{1-x}Cr_x$. Quantum elastodynamic calculations predict surface acoustic wave (SAW) peak splitting arising from elastic stiffening of δ-ferrite during spinodal decomposition. Classical first-principles elastodynamics also quantitatively calculate Rayleigh SAW peak splitting arising from spinodal stiffening as a non-destructive indicator of embrittlement, in agreement with experiment~~and beyond what classical molecular dynamics captures without electronic structure~~. More broadly, ~~the same~~ similar SAW-based stiffening ~~signature may extend to~~ signatures may prove useful for detecting other embrittlement mechanisms in structural alloys~~, such as Cu precipitation in reactor pressure vessel (RPV) steels~~.

Keywords: Stainless steel, binary alloy, iron-chrome, spinodal decomposition, surface acoustic wave, Ab initio calculation, stiffening, embrittlement, thermal diffusivity, δ–ferrite, quantum elastodynamics (QED)

Highlights

- ~~Spinodal decomposition in bcc~~ Ab initio simulations predict stiffening during spinodal decomposition in δ–ferrite ~~induces stiffening captured by ab initio quantum elastodynamics~~
- Composition-dependent elasticity maps to ~~Rayleigh~~ surface acoustic wave frequency shifts ~~and peak splitting~~
- ~~Theoretical and experimental agreement demonstrates stiffening as a non-destructive indicator of~~ Theory and experiment agree: stiffening non-destructively indicates embrittlement

1. Introduction and Background

Recent structure–property studies have linked spinodal decomposition to stiffening of bcc δ–ferrite in cast austenitic stainless steels (CASS), and have shown that this stiffening can be detected through picosecond ultrasonic surface acoustic wave (SAW) peak splitting [1–3]. This paper establishes a quantitative theoretical foundation for predicting that SAW signature from first principles by combining elastodynamics with density functional theory (DFT) to connect composition, elastic response, and Rayleigh-wave observables.

For high–melting-point metals (including bcc δ–ferrite), elasticity is often closely tied to plastic response: dislocation energetics are largely stored in the Hookean lattice outside the dislocation core, and thermally activated processes during deformation can modify elastic properties [4]. More broadly, cohesive properties quantified by lattice elasticity correlate with melting point through their shared dependence on interatomic bonding and electronic structure [5–9]. In ferromagnetic Fe–Cr alloys, these quantities vary strongly with concentration [7], and the experimentally observed SAW peak splitting upon spinodal decomposition is

consistent with an anomalous concentration dependence of elastic constants associated with electronic-structure effects (including Fermi-surface topology changes [8]).

During spinodal decomposition of bcc δ–ferrite, chromium partitioning produces Fe-rich α and Cr-rich α' regions whose differing compositions imply differing stiffness. As a result, elastic stiffening can serve as a mechanistically grounded proxy for embrittlement in stainless steels undergoing thermal and/or irradiation-assisted aging [2,3,10–14]. Because the Rayleigh SAW frequency depends directly on elastic moduli, picosecond ultrasonic pump–probe spectroscopy provides a direct, non-destructive route to infer stiffness changes and therefore track the progression toward embrittlement [12].

Motivated by these links, we develop a thermodynamics-to-observable workflow for bcc $Fe_{1-x}Cr_x$ as a model for δ–ferrite: (i) we identify spinodally unstable composition windows using the curvature of the demixing free energy, (ii) we quantify the composition dependence of cubic elastic constants from DFT, and (iii) we map these elastic inputs into Rayleigh SAW frequency ranges using elastodynamic constraints relevant to picosecond ultrasonics. Finally, we separate a solid-solution trend $f_{SAW}(x_{Cr})$ from an evolution trend $f_{SAW}(\phi)$ parameterized by the spinodal conversion fraction ϕ from ~~molecular dynamics (MD)~~our earlier atomistic simulations of decomposed Fe–Cr [1], enabling direct interpretation of SAW peak shifts and splitting in terms of composition and decomposition state.

2. Methods

2.1. Elastodynamic Methods

~~In this work, materials are modeled as anisotropic unbounded media through which elastic waves propagate according to~~ Elastic waves in each homogeneous alloy obey the elastodynamic wave equation [10]:

$$\rho \frac{\partial^2 u_i}{\partial t^2} = C_{ijkl} \frac{\partial^2 u_k}{\partial x_j \partial x_l} \tag{1}$$

where C_{ijkl} are fourth-order stiffness tensor entries ~~that are permutatively~~ indexed by $i, j, k, l = 1, 2, 3$, ρ is density, and ~~U is the wave function.~~ u is the displacement field. On its own, Eq. 1 describes bulk waves in an unbounded medium. The SAW measured by transient grating spectroscopy (TGS), however, is a Rayleigh wave guided by the free surface. It is obtained from Eq. 1 for a half-space $x_3 \geq 0$ by requiring a traction-free surface, $\sigma_{i3} = C_{i3kl}\,\partial u_k/\partial x_l = 0$ at $x_3 = 0$, and displacements that decay with depth. Because crystal orientation is not resolved in the present calculations, we solve this surface problem for the isotropic Voigt–Reuss–Hill (VRH) average of each elastic tensor (Sec. 2.4). The Rayleigh velocity is then $v_R = v_s\sqrt{q}$, where q is the root in $0 < q < 1$ of

$$q^3 - 8q^2 + (24 - 16\beta)q - 16(1 - \beta) = 0, \qquad \beta = v_s^2/v_p^2, \tag{2}$$

with $v_s = \sqrt{G/\rho} = \sqrt{E/[2(1+\nu)\rho]}$ and $v_p = \sqrt{(B + 4G/3)/\rho}$. In practice we evaluate $v_R = (0.874 + 0.196\nu - 0.042\nu^2 - 0.055\nu^3)\,v$ which reproduces the root of Eq. 2 to within 0.02% for $0.20 \leq \nu \leq 0.35$. The SAW frequency is $f_{SAW} = v_R/\lambda$ with $\lambda = 6.4\,\mu$m, the grating period of our TGS measurements [1]. Density is interpolated linearly between bcc Fe (7870 kg m^{-3}) and Cr (7200 kg m^{-3}).

The elastic tensor is calculated using a variable cell relaxation algorithm that consists of a loop over iterative ~~SCF~~ self-consistent field (SCF) calculations while adjusting atomic positions and lattice constants. Elasticity theory relates the strain applied to the unit cell to the resulting calculated stress as an inverse problem, which could be solved using a non-linear solver with prior data as an initial guess vector. Deterministic thermoelastic relationships are also used to calculate elastic tensors from changes in total energy and volume using ab initio stress calculations. Both methods are used in a self-consistent fashion to converge to a solution that solves the elasticity inverse problem, $\underline{C}\vec{\epsilon} = \vec{\sigma}$, as follows [15,16]:

$$\begin{bmatrix} C_{1111} & C_{1122} & C_{1133} & C_{1123} & C_{1113} & C_{1112} \\ & C_{2222} & C_{2233} & C_{2223} & C_{2213} & C_{2212} \\ & & C_{3333} & C_{3323} & C_{3313} & C_{3312} \\ & & & C_{2323} & C_{2313} & C_{2312} \\ & \textit{symm} & & & C_{1313} & C_{1312} \\ & & & & & C_{1212} \end{bmatrix} \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ 2\epsilon_{23} \\ 2\epsilon_{13} \\ 2\epsilon_{12} \end{bmatrix} = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix} \tag{3}$$

or in Voigt notation:

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ & & C_{33} & C_{34} & C_{35} & C_{36} \\ & & & C_{44} & C_{45} & C_{46} \\ & \textit{symm} & & & C_{55} & C_{56} \\ & & & & & C_{66} \end{bmatrix} \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \\ \epsilon_6 \end{bmatrix} = \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix} \tag{4}$$

2.2. DFT Methods

Material properties are calculated from a self-consistent, non-linear field calculation ~~,~~ (Fig.~~S1) which is~~ S1), a variational method that optimizes ~~over charge density that~~ the charge density and adjusts atomic positions until a threshold in energy minimization is achieved in a simplified ion-core model of atoms and electrons ~~in a given material to solve the many-body wave equation [17]. Electron dispersion is calculated using a non-consistent field calculation to interpolate the energy calculations in momentum space. The phonon dispersion is calculated from the dynamical matrix obtained from ab initio calculations [18,19].~~ [17] .

Self-consistent, non-linear field calculation algorithms calculate external potentials with a Poisson equation solver, (Fig. S1) which depends on simulation geometry, atomic positions of ions, pseudopotentials used for the atoms involved, and other inputs. This external potential is then used as a guess for the electron density with a Schrödinger equation solver [19], which is then used as inputs to a self-consistent field loop that calculates the charge density iteratively to minimize the total energy. The accuracy of this algorithm depends on the cut-off applied to the energy minimization, the charge density, the size of the momentum grid, and other convergence thresholds on the outputs [17–19].

In order to model the BCC Fe–Cr δ–ferrite solid solution, the ~~sequential~~ special quasirandom structure (SQS) methodology was employed [20]. The mcsqs software package within the Alloy Theoretic Automated Toolkit (ATAT) was used in order to generate SQS supercells. The Vienna ab initio Simulation Package (VASP) [21] was used to perform the density functional theory (DFT) total energy calculations, which were used to derive the elastic properties as well as the enthalpy of mixing. Spin–polarized DFT calculations were performed on 40–atom SQS supercells employing Γ–point sampling of reciprocal space along with a 500 eV planewave cutoff energy. The Projector–Augmented Wave (PAW) method [22] was employed for the ionic pseudopotential, and the ~~Perdew–Burke–Ernzenhoffer parameterization~~ Perdew–Burke–Ernzerhof parameterization [23] of the Generalized Gradient Approximation (GGA) of the exchange–correlation functional. ~~Classical molecular dynamics is used only to parameterize the~~ Calculations were performed at $x_{Cr} = 0$, 0.20, 0.50, 0.95, and 1.00.

2.3. Atomistic simulations of decomposed Fe–Cr

The spinodal conversion fraction ϕ ~~; it does not reproduce the observed SAW frequency shifts without electronic-structure-informed elasticity (Fig. 3b)~~and the corresponding elastic moduli are taken from our earlier atomistic study [1] (Sec. 3.7 and Fig. 8 therein). These simulations were performed with LAMMPS [24], not VASP, and include no electronic structure. A $50 \times 50 \times 50$ bcc cell (lattice parameter 2.866 Å) of 250,000 atoms with equiatomic composition was populated randomly and phase-separated by Monte Carlo atom swaps at 10 K (3000 attempts every 50 steps for 40,~~underscoring the necessity of first-principles calculations.~~

2.4. ~~Analysis of DFT Results~~

~~With the elastic tensor obtained from quantum mechanical calculations,elastic moduli could be calculated. The significance of elastic moduli in characterizing material behavior is highlighted, and the~~ 000 steps). The swaps were driven by a structure-generation pair interaction only: zero for like pairs and a strongly repulsive nearest-neighbour Fe–Cr Lennard-Jones term (ϵ = 10 eV). Intermediate states were produced by converting a fraction (10–50%) of the Fe atoms to Cr and the same fraction of Cr atoms to Fe. After box and ionic relaxation, the elastic constants of ~~iron-chrome with increasing chromium content are reported using density functional theory. By considering embrittlement criteria,~~ each state were calculated by finite deformation (strain 10^{-6}) with the ~~propensity of materials to undergo embrittlement could be evaluated to assess their mechanical stability.~~

~~Thermal properties are described by the Debye model and a lower limit for the thermal conductivity was derived at high temperatures, such that [25,26]: where κ is thermal conductivity, θ_D~~ Fe–Cr two-band embedded-atom-method (EAM) potential of Bonny et al. [27], for ten alloys generated with different random seeds. On this scale, ϕ = 0.5 is the ~~Debye temperature, C_V is the heat capacity at constant volume, v_m is the average acoustic velocity, A is the phonon mean free path, m is the number of atoms, M is the mass, ρ is the density, E is the elastic Young's modulus, and A, B are correlation constants approximated from the relationship between the mean acoustic velocity and ratio of $\sqrt{\frac{E}{\rho}}$, which is valid for a wide range of Poisson ratios without loss of accuracy. The value for A used for these constants are taken as 0.87 ± 0.02. Other studies calculate thermal conductivity using the Einstein model, which enables calculating thermal properties as a function of temperature, such that: $\kappa_{min}(T) = (\frac{\pi}{6})^{\frac{1}{3}} k_B n^{\frac{2}{3}} \sum_i v_i (\frac{T}{\Theta_i})^2 \int_0^{\frac{\Theta_i}{T}} dx \frac{x^3 e^x}{(e^x-1)^2}$ where $\Theta_i = v_i (\frac{\hbar}{k_B})(6\pi^2 n)^{\frac{1}{3}}$ [28].~~ random homogeneous alloy and ϕ = 1 the fully separated structure. Relaxed cell volumes were not retained, so the density is held at the equiatomic value of 7535 kg m^{-3} when these moduli are mapped to f_{SAW} with Eq. 2. All LAMMPS inputs, random seeds, and potential files are provided in the accompanying repository. For direct comparison with the homogeneous-alloy DFT and literature data, we also computed random (homogeneous) bcc alloys at x_{Cr} = 0, 0.20, 0.50, 0.95, and 1.00 with the same stiffness procedure, cell, and potential, using ten random seeds for each mixed composition (one for the pure elements), and at x_{Cr} = 0.10, 0.30, 0.36, 0.40, 0.60, 0.70, and 0.80 with three seeds each. The random $Fe_{50}Cr_{50}$ alloy reproduces the published value (C_{11} = 252.9 ± 0.1 versus 252.7 ± 0.1 GPa), and the seed-to-seed scatter in C_{11} is at most 0.25 GPa. The decomposition procedure was repeated in a $25 \times 25 \times 25$ cell (31,250 atoms) at twelve compositions, x_{Cr} = 0.05–0.95, for five alloys (random seeds) at each; we report the mean over the five alloys, and the seed-to-seed standard deviation of the stiffening on decomposition is at most 0.1 percentage points. These runs use the current NIST release of the Bonny et al. potential files, which are numerically identical to those used in Ref. [1] apart from two format corrections required by current LAMMPS versions: zero-filled pair tables in the s-band file and finite values at r = 0.

To extend the decomposition series beyond the equiatomic alloy, we repeated the full procedure at x_{Cr} = 0.20, 0.36, and 0.95 in a $25 \times 25 \times 25$ cell (31,250 atoms) with the same number of attempted swaps per atom. Away from 50/50, converting equal fractions of the Fe and Cr atoms changes the overall composition. We therefore converted a fraction c of the Cr atoms and a fraction cN_{Cr}/N_{Fe} of the Fe atoms, with $c = 2(1 - x_{Cr})(1 - \phi)$; this preserves the composition (to within 0.004 in all structures) and is identical to the original procedure at 50/50. One alloy was generated per composition. As a check, the same small-cell procedure at 50/50 reproduces C_{11} of the original series to within 0.8 GPa at every ϕ.

2.4. Analysis of DFT Results

With the elastic tensor obtained from quantum mechanical calculations, the isotropic VRH moduli used in Eq. 2 are calculated as follows; the same moduli enter the Pugh embrittlement criterion and the thermal estimates reported in the Supplementary Material.

$$
\begin{aligned}
C' &= \frac{C_{11} - C_{12}}{2}, & B &= \frac{C_{11} + 2C_{12}}{3}, \\
C_{11} &= \frac{3B + 4C'}{3}, & C_{12} &= C_{11} - 2C', \\
G_V &= \frac{C_{11} - C_{12} + 3C_{44}}{5}, & G_R &= \frac{5(C_{11} - C_{12})C_{44}}{4C_{44} + 3(C_{11} - C_{12})}, \\
G &= \frac{G_V + G_R}{2}, & E &= \frac{9BG}{3B + G}, & \nu &= \frac{3B - E}{6B}.
\end{aligned}
\tag{5}
$$

Equation 5 gives the cubic forms used for the literature tensors. For the DFT points, B and G are VRH averages of the full calculated tensor, because the SQS cells are not exactly cubic. All figures are generated from tabulated inputs by scripts provided in the accompanying repository (see Data Availability).

3. Results

3.1. Spinodal stability from demixing free-energy curvature

The local stability of a binary solid solution is governed by the curvature of ~~the demixing free energy $\Delta G(x_{Cr})$~~ its free energy of mixing with respect to composition. ~~A negative second derivative $\partial^2 \Delta G / \partial x_{Cr}^2 < 0$ identifies a spinodally unstable region:~~ Where $\partial^2 \Delta G_{mix} / \partial x_{Cr}^2 < 0$, infinitesimal composition fluctuations lower ~~ΔG~~the free energy, and demixing proceeds spontaneously without the need for nucleation barriers. ~~This curvature criterion provides a thermodynamically rigorous definition of the composition window in which microstructural evolution is inevitable once sufficient atomic mobility is available~~Figure 1 is evaluated at 298.15 K from the concentration-dependent interaction models that Levesque et al. fitted to PWSCF–PAW and SIESTA–NC calculations [29], $\Omega(x) = (x - a)(bx^2 + cx + d)$, together with the ideal configurational entropy S_{mix}. Following the sign convention of that model, we plot the demixing free energy

$$
\Delta G_{demix} = \Omega(x)\, x(1 - x) + T S_{mix} = -\Delta G_{mix}. \tag{6}
$$

The spinodal criterion $\partial^2 \Delta G_{mix} / \partial x_{Cr}^2 < 0$ therefore appears as positive curvature, $\partial^2 \Delta G_{demix} / \partial x_{Cr}^2 > 0$, in Fig. 1b. Figure 1b compares three estimates of this curvature for the PWSCF–PAW model: the exact analytic second derivative, three-point numerical second differences of its tabulated values on the non-uniform composition grid, and a cruder two-point scheme together with a quadratic fit to its values. The three-point values fall on the exact curve. The two-point scheme distorts the magnitudes on the uneven grid, most strongly at $x_{Cr} = 0.70$, next to a duplicated composition in the tabulation, but agrees on where the instability begins.

Figure 1b shows that the ~~onset of negative curvature occurs over a finite range of chromium compositions relevant to δ-ferrite in stainless steels. This thermodynamic instability defines the regime in which~~ homogeneous alloy is unstable for $0.21 \lesssim x_{Cr} \lesssim 0.75$ in the PWSCF–PAW model (with a second, narrow window at 0.87–0.98); the SIESTA–NC model gives $0.25 \lesssim x_{Cr} \lesssim 0.99$. Both windows contain the Fe–36Cr alloy in which spinodal decomposition was recently detected through TGS stiffening [33]. Because the entropic term lowers the curvature in proportion to T, these windows narrow at the higher temperatures used for aging. At 748 K (475 °C) the CALPHAD assessment of Xiong et al., validated by heat capacity, Curie temperatures and atom-probe tomography, places the chemical spinodal at $x_{Cr} \approx 0.30$–0.71 within a miscibility gap of ≈ 0.14–0.88 [34]. Fe–36Cr therefore lies inside the chemical spinodal at the aging temperature, close to its Fe-rich edge, consistent with their observation at 773 K that 37.8 at.% Cr decomposes spinodally whereas 26.7 at.% Cr decomposes by nucleation [34]. Within them, elastic and acoustic properties cannot be interpreted assuming a homogeneous solid solution. Instead, any observable elastic response must be understood as emerging from a system that is intrinsically driven toward compositional modulation, providing the foundation for linking free-energy curvature to stiffness evolution and ultimately to ultrasonic observables.

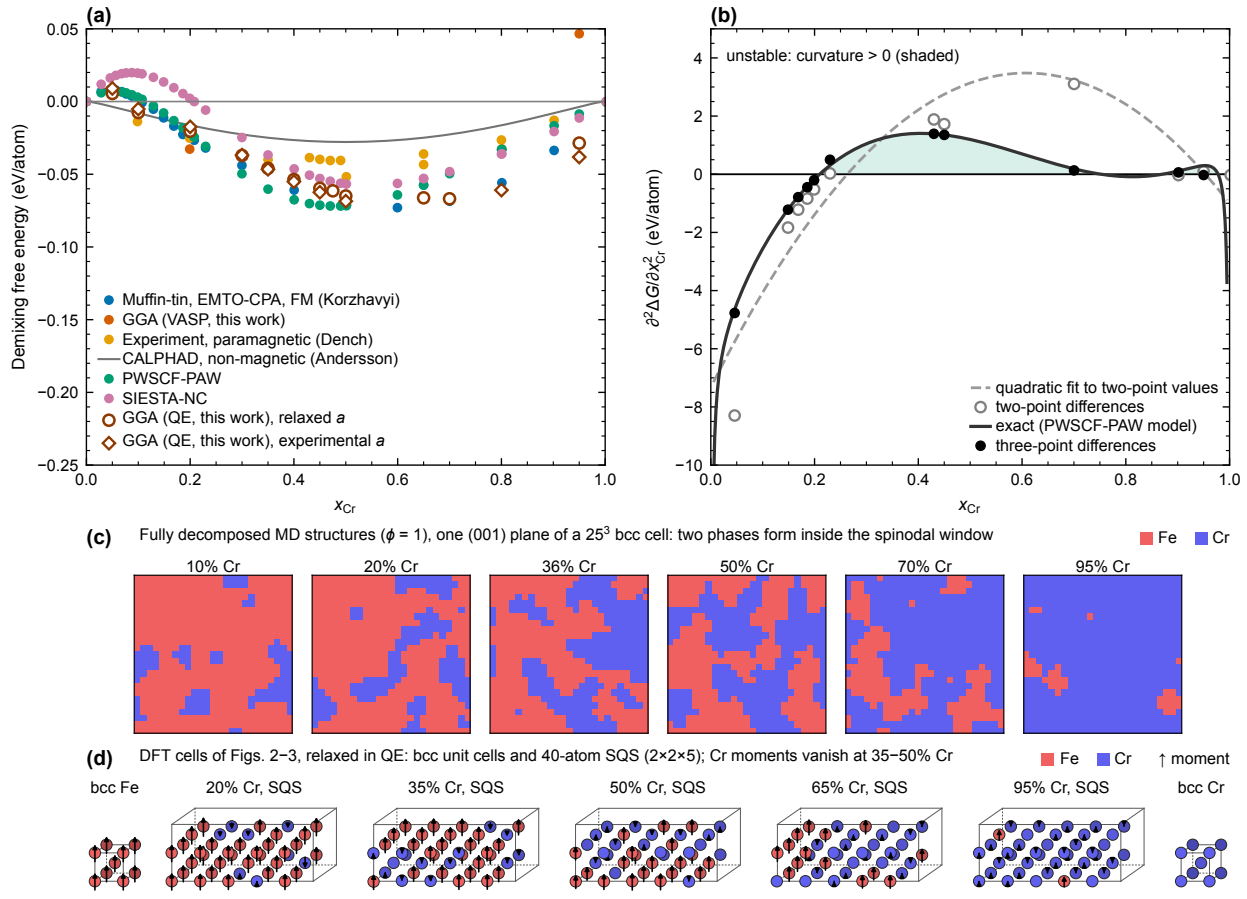


Figure 1: (a) Demixing free energy ~~ΔG~~ $\Delta G_{\mathrm{demix}} = -\Delta G_{\mathrm{mix}}$ (Eq. 6) versus chromium composition x_{Cr} at 298.15 K: muffin-tin (EMTO-CPA, ferromagnetic) enthalpies of Korzhavyi et al. [7], present GGA calculations with VASP (filled) and Quantum ESPRESSO (40-atom SQS, references FM Fe and AFM Cr; open circles at the PBE-relaxed lattice parameter, open diamonds at the room-temperature experimental lattice parameter with the references at theirs, which changes $\Delta G_{\mathrm{demix}}$ by at most 10 meV/atom), calorimetric enthalpies of Dench [30] (measured near 1600 K relative to paramagnetic Fe and Cr), the CALPHAD bcc description of Andersson and Sundman [31] (non-magnetic part, line), and the PWSCF–PAW and SIESTA–NC interaction models of Levesque et al. [29]. All series include the ideal configurational entropy as in Eq. 6; the measured vibrational entropy of mixing, about $0.2\,k_{\mathrm{B}}$/atom [32], would change $\Delta G_{\mathrm{demix}}$ by at most 5 meV/atom at this temperature. (b) Curvature ~~$\partial^2\Delta G/\partial x_{\mathrm{Cr}}^2$, identifying spinodally unstable~~ of the PWSCF–PAW demixing free energy. Solid line: exact analytic second derivative of the model. Filled circles: three-point second differences of its tabulated values on the non-uniform composition ~~windows where~~ grid. Open circles: two-point second differences of the same values, with a quadratic least-squares fit (dashed). Shading marks positive demixing curvature~~becomes~~, i.e., negative mixing curvature, which identifies spinodally unstable compositions. (c) One (001) atomic plane of fully decomposed (ϕ = 1) MD structures (25^3 bcc cells; Sec. 2.3) at the indicated Cr contents; red, Fe; blue, Cr. Inside the spinodal window the alloy separates into interpenetrating Fe-rich α and Cr-rich α' domains, whereas at the dilute ends the minority element forms only isolated clusters. (d) The DFT cells behind the first-principles tensors of Figs. 2 and 3: bcc unit cells of Fe and Cr and 40-atom SQS ($2\times2\times5$ conventional cells), relaxed in Quantum ESPRESSO, with their local magnetic moments (arrows; red, Fe; blue, Cr). The Cr moments, antiparallel to Fe at 20% Cr, vanish between 35 and 50% Cr. To be updated: QE points at 80 at.-pct Cr (PBE-relaxed) and at 65 at.-pct Cr (experimental lattice parameter) will be added when the Engaging runs finish.

3.2. Elastic stiffening and elastodynamic mapping to Rayleigh SAW observables

To connect thermodynamic instability to experimentally measurable quantities, we compute the cubic elastic tensor across x_{Cr} and propagate these elastic inputs through ~~elastodynamic constraints relevant to surface acoustic waves.In the present geometry,~~ the surface boundary-value problem of Sec. 2.1. Although f_{SAW} depends on the full tensor through B and G, we use the longitudinal stiffness ~~component~~ C_{11} ~~plays~~

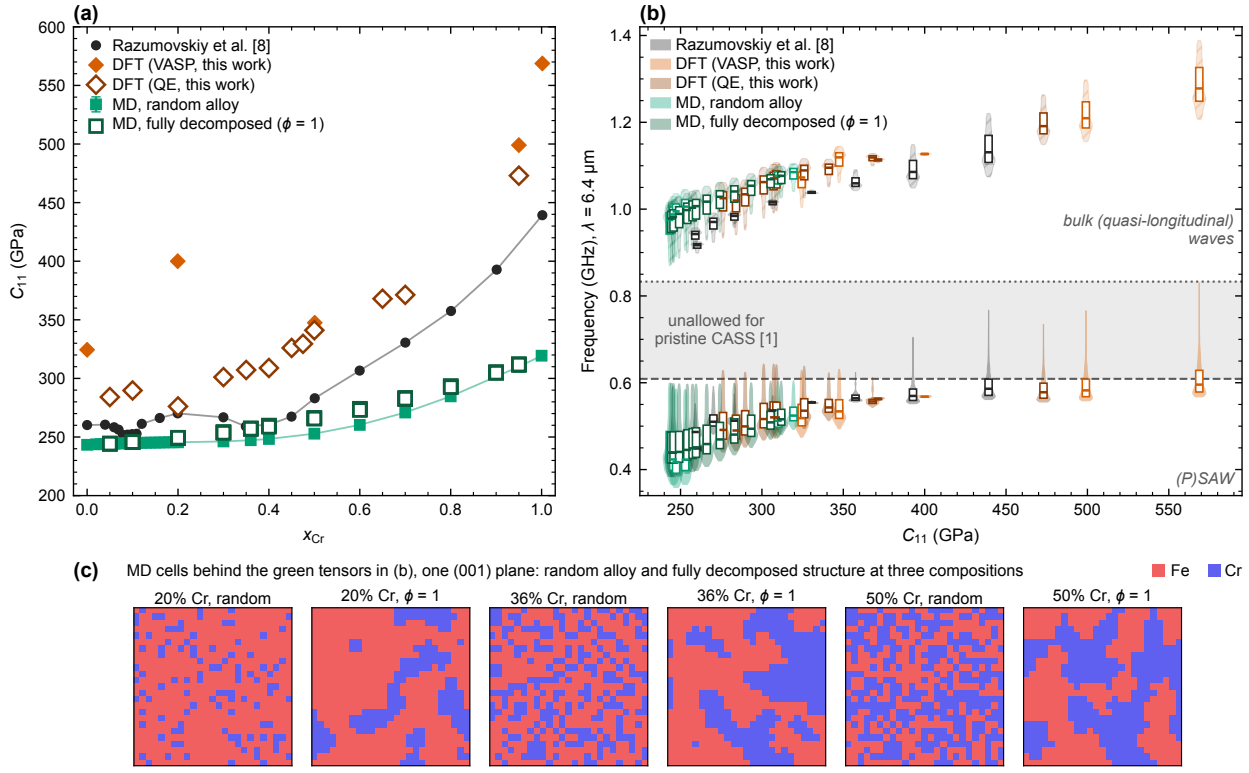


Figure 2: (a) ~~Composition dependence~~ C_{11} of ~~the bcc~~ bcc $Fe_{1-x}Cr_x$ ~~cubic elastic tensor~~: literature calculations of Razumovskiy et al. [8] (black circles; reconstructed from digitized C', ~~shown for~~ C_{44}, and B), present DFT calculations on 40-atom SQS with VASP (filled red diamonds) and Quantum ESPRESSO (open diamonds; SM S3), MD (EAM) random alloys (green squares; mean ± standard deviation of ten alloys at the ~~longitudinal stiffness~~ C_{11} DFT compositions and three elsewhere; Sec. 2.3), and fully decomposed MD structures ($\phi = 1$, open squares; 25^3 cells). (b) ~~Elastodynamic mapping from elastic stiffness to Rayleigh surface~~ Orientation-resolved acoustic ~~wave frequency~~ f_{SAW} frequencies at $\lambda = 6.4\,\mu$m, ~~indicating elastodynamically allowed~~ plotted at the C_{11} of each tensor in (a). For each tensor, 10,000 random grain orientations were computed with the surface Green's-function method of Du and Zhao [10], as in Ref. [1]. Lower violins: (pseudo-)SAW frequencies (pseudo-SAW retained when its intensity is at least 2×10^{-3} of the SAW). Upper violins: quasi-longitudinal bulk waves along the same propagation directions. Boxes mark the interquartile range and median. The shaded band is the elastodynamically unallowed range of pristine 304-type CASS [1], from its maximum (pseudo-)SAW frequency ~~ranges for a homogeneous solid solution~~ (0.609 GHz, dashed) to its bulk-wave frequency (0.833 GHz, dotted). (c) One (001) atomic plane of the MD cells behind the green tensors: random alloy and fully decomposed structure at $x_{Cr} = 0.20$, 0.36, and 0.50 (red, Fe; blue, Cr). To be updated: the QE point at 80 at.-pct Cr will be added; the QE set at the experimental lattice parameter is in SM Section S4.

~~a dominant role in controlling the Rayleigh SAW frequency through the dispersion relation, making it a sensitive indicator~~ as a compact index of composition-dependent stiffening ~~. Small overpredictions in absolute frequency are attributed to finite-size SQS representations and the absence of magnetic and chemical short-range order beyond the supercell length scale. Importantly, relative stiffening trends~~ (Fig. 2a). The black circles are literature calculations by Razumovskiy et al. [8], the red diamonds are ours, and the green squares are MD (EAM) random alloys computed here with the procedure of our earlier study [1]; the open squares are the corresponding fully decomposed structures, which are stiffer than the random alloy at every composition, most strongly at intermediate compositions (e.g., 257.4 versus 247.3 GPa at $x_{Cr} = 0.36$). The present DFT tensors are systematically stiffer than the reference data (e.g., $C_{11} = 324$ versus 260 GPa for bcc Fe) and correspondingly predict higher frequencies. Possible contributors include the Γ-point-only k-point sampling, the finite SQS size, and the magnetic configuration. We therefore interpret relative trends rather than absolute frequencies. Independent Quantum ESPRESSO calculations on 40-atom SQS with converged Brillouin-zone sampling (open diamonds; SM S3) agree with the literature at $x_{Cr} = 0.20$ and 0.95

(f_{SAW} = 0.509 and ~~peak splitting remain robust, demonstrating that first-principles elasticity captures the essential physics governing SAW observables~~0.591 GHz versus 0.512 and 0.590 GHz) but rise steadily in between, without the minimum near 36% Cr.

As shown in Fig. 2, C_{11} ~~exhibits a systematic dependence~~ depends systematically on chromium concentration (non-monotonically in the literature and DFT data), producing corresponding shifts in the ~~elastodynamically allowed~~ Rayleigh SAW frequency~~range. Importantly, this mapping provides a quantitative bridge between first-principles elasticity and experimentally observed SAW peak positions.~~ . For a homogeneous solid solution, each composition corresponds to a single tensor, and hence to a single isotropic f_{SAW} but a distribution of (P)SAW frequencies over grain orientations. In our earlier study, elastodynamic calculations for pristine CASS over randomly sampled grain orientations defined an "unallowed" frequency range, above the highest (pseudo-)SAW frequency and below the bulk-wave frequency, in which no surface mode of the pristine material exists; (P)SAW peaks observed there after aging were attributed to stiffened δ-ferrite [1]. Figure 2b repeats that calculation for every Fe–Cr tensor (10,000 grain orientations each, $\lambda = 6.4\,\mu$m) [10] . The bulk waves of all tensors lie above the range ($\geq$ 0.87 GHz). In the (P)SAW branch, no literature tensor up to $x_{Cr} = 0.7$ and none of the MD tensors reaches it. Full decomposition raises the highest MD (P)SAW frequency, most at intermediate compositions (from 585 to 596 MHz at $x_{Cr} = 0.36$), but not into the range. Only the stiffest tensors enter it: Cr-rich literature alloys (9% and 29% of grains above 0.609 GHz at $x_{Cr} = 0.9$ and 1.0) and the present DFT tensors (30% and 46% at $x_{Cr} = 0.95$ and 1.0; their systematic overstiffening also places 6–8% of grains of Fe and $Fe_{50}Cr_{50}$ in the range). Frequencies in this range therefore indicate stiffness beyond that of a homogeneous solid solution of δ-ferrite or Fe–36Cr composition, consistent with the stiffer Cr-rich α' regions formed by spinodal decomposition (Fig. 3a). Within the spinodally unstable composition window, elastic stiffening therefore translates ~~directly~~ into measurable acoustic signatures, enabling stiffness changes to be inferred non-destructively.

3.3. Separating composition trends from spinodal conversion trends

While elastic constants vary with chromium concentration even in a homogeneous solid solution, spinodal decomposition introduces an additional, time-dependent microstructural evolution. To disentangle these effects, it is necessary to distinguish changes driven purely by composition from those arising from the progression of phase separation. We therefore report two complementary relationships that isolate these contributions within a unified elastodynamic framework.

Figure 3~~demonstrates that while~~ a shows how chromium partitioning produces stiffening. The Fe–36Cr composition lies near a local minimum of the homogeneous $f_{SAW}(x_{Cr})$ curve. Spinodal decomposition moves material toward Fe-rich α and Cr-rich α' compositions (arrows); in both directions the homogeneous-alloy frequency increases. The DFT set, with only five compositions, is consistent with this non-monotonic dependence but cannot resolve the minimum. The MD random alloys, an independent classical model, also show a frequency minimum, at $x_{Cr} \approx 0.36$–0.40, i.e. at the Fe–36Cr composition, even though their C_{11} increases monotonically (Fig. 2a): the minimum comes from the shear moduli, which control the Rayleigh velocity (C_{44} decreases from 116 GPa for Fe to 106 GPa at $x_{Cr} = 0.5$). C_{11} is therefore a convenient index of stiffening, but the SAW response is governed by shear stiffness. Figure 3a describes the properties of the constituents; the effective response of the two-phase microstructure additionally depends on phase fractions and on the domain size relative to the SAW wavelength.

Figure 3b isolates the effect of decomposition at fixed overall composition ($Fe_{50}Cr_{50}$). In the MD (EAM) simulations, f_{SAW} increases monotonically from 0.469 GHz for the random alloy ($\phi = 0.5$) to 0.490 GHz for the fully separated structure ($\phi = 1$), an increase of about 4.6% under the fixed-density assumption. The empirical-potential and first-principles results therefore agree in sign. Repeating the decomposition at twelve compositions from $x_{Cr} = 0.05$ to 0.95 (Sec. 2.3) gives frequency increases between the random and fully separated states that are nearly symmetric about the equiatomic composition: 0.5%, 1.2%, 2.7%, 4.3%, 5.0%, 5.4%, and 5.7% for $x_{Cr} = 0.05$, 0.10, 0.20, 0.30, 0.36, 0.40, and 0.50, and 5.2%, 4.1%, 2.7%, 1.1%, and 0.4% for $x_{Cr} = 0.60$, 0.70, 0.80, 0.90, and 0.95 (means over five alloys; standard deviations at most 0.1 percentage points). Stiffening on decomposition is therefore largest within the spinodally unstable window of Fig. 1b, including the Fe–36Cr composition, and nearly vanishes for the dilute alloys at either end. The fully separated state is an idealized limit, so these values bound the stiffening from above. At

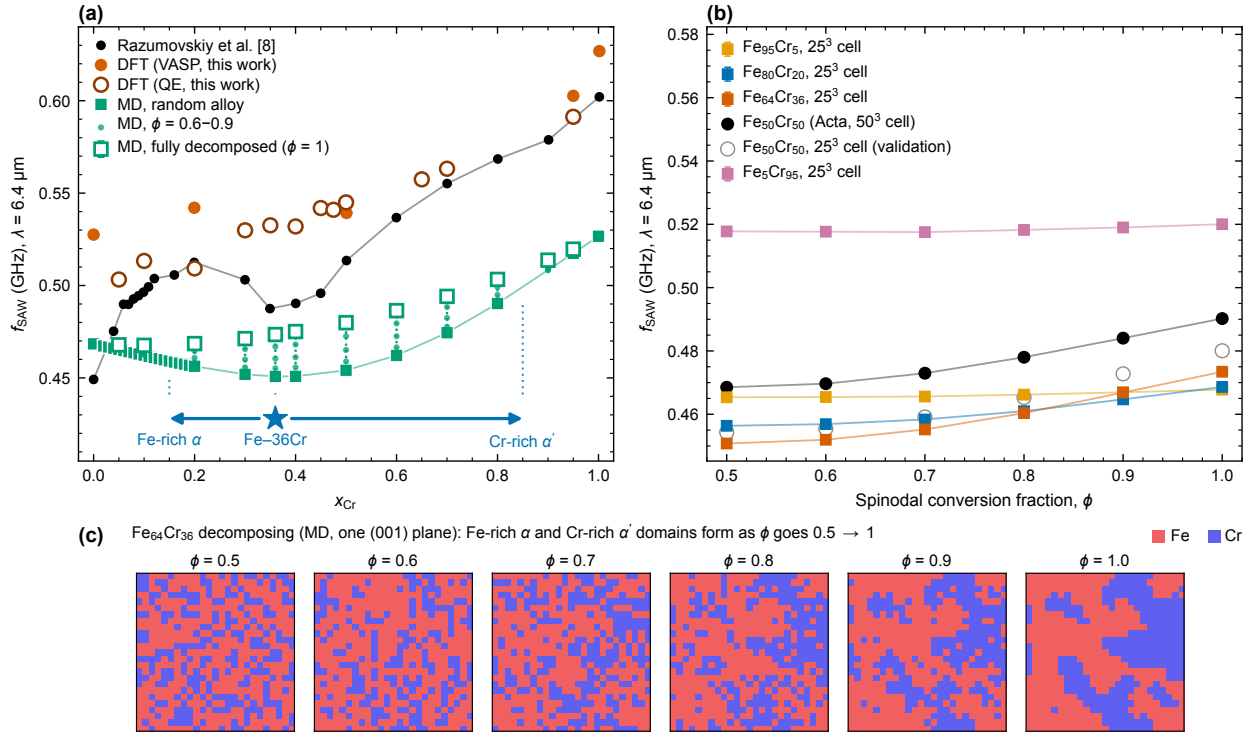


Figure 3: (a) Rayleigh SAW frequency f_{SAW} ~~as a function of~~ versus chromium composition x_{Cr} ~~using DFT-derived elastic constants (solid-solution trend~~ $\lambda = 6.4\,\mu$m) for the literature (black), present DFT (VASP, filled red; Quantum ESPRESSO, open), and MD random-alloy (green) data of Fig. 2. At each MD composition, small dots (~~b~~$\phi = 0.6$–0.9) ~~Rayleigh SAW~~ and an open square ($\phi = 1$) show the frequency ~~as a function~~ of the decomposing structure, each series anchored to the random-alloy value at that composition, so the ladder height is the stiffening on decomposition (as in b). Below the data, the star marks Fe–36Cr [33] and the arrows indicate schematically the partitioning toward Fe-rich α and Cr-rich α' during spinodal decomposition; dotted lines rise from each composition toward the data. Arrow endpoints are illustrative, not measured phase compositions. (b) f_{SAW} versus spinodal conversion fraction ϕ from ~~molecular dynamics, isolating microstructural evolution effects during~~ the MD (EAM) decomposition simulations (Sec. 2.3): equiatomic $Fe_{50}Cr_{50}$ from Ref. [1] (black, 50^3 cell, fixed density 7535 kg m^{-3}), and $Fe_{95}Cr_5$, $Fe_{80}Cr_{20}$, $Fe_{64}Cr_{36}$, and Fe_5Cr_{95} from the present 25^3-cell calculations (squares); open circles repeat $Fe_{50}Cr_{50}$ in the 25^3 cell as a check. $\phi = 0.5$ is the random alloy and $\phi = 1$ the fully separated structure. (c) One (001) atomic plane of the 25^3 MD cell of $Fe_{64}Cr_{36}$ from $\phi = 0.5$ to 1.0 (red, Fe; blue, Cr): Fe-rich α and Cr-rich α' domains form and coarsen as ϕ increases; these are the structures whose frequencies are plotted in (b). All 25^3-cell values are means over five MD alloys (random seeds); their ±1 s.d. error bars (at most 0.5 MHz) are smaller than the symbols. To be updated: the QE point at 80 at.-pct Cr will be added; the QE set at the experimental lattice parameter is in SM Section S4.

50/50, the small-cell series lies about 2% below the original series in absolute frequency because the original moduli used a different isotropic average; the relative changes agree to within about one percentage point. Together, $f_{SAW}(x_{Cr})$ captures the baseline elastic response of a homogeneous alloy, ~~the evolution of~~ and $f_{SAW}(\phi)$ reflects the additional stiffening associated with progressive spinodal conversion. This separation enables experimental SAW measurements to be interpreted in terms of both composition and decomposition state, providing a pathway to distinguish early-stage spinodal evolution from equilibrium compositional effects.

4. Discussion

Taken together, the results establish a closed thermodynamics-to-observable pathway for interpreting picosecond ultrasonic measurements in δ–ferrite–containing stainless steels. The free-energy curvature criterion identifies the composition window in which the homogeneous solid solution is intrinsically unstable,

defining the regime where elastic and acoustic properties must reflect ongoing or incipient spinodal decomposition rather than equilibrium alloy behavior.

Within this thermodynamically unstable regime, first-principles elasticity reveals systematic stiffening trends that are propagated through elastodynamic constraints into Rayleigh SAW frequency shifts. The direct mapping between elastic constants and SAW observables provides a quantitative basis for interpreting peak shifts and splitting as signatures of stiffness evolution, rather than purely phenomenological indicators.

Crucially, separating the solid-solution trend $f_{\mathrm{SAW}}(x_{\mathrm{Cr}})$ from the spinodal conversion trend $f_{\mathrm{SAW}}(\phi)$ resolves a key ambiguity in experimental interpretation. While composition alone can shift SAW frequencies, progressive spinodal decomposition introduces additional stiffening that cannot be captured by composition-dependent elasticity alone. The combined framework therefore enables SAW peak splitting to be interpreted as a mechanistically grounded indicator of microstructural evolution toward embrittlement, rather than as a purely compositional effect.

This separation clarifies why SAW peak splitting emerges as a robust, non-destructive diagnostic of spinodal decomposition in CASS and related alloys, and suggests a general strategy for extending ultrasonic embrittlement detection to other phase-separating structural materials.

5. Conclusions

We presented a compact thermodynamics-to-elastodynamics workflow for bcc Fe–Cr δ–ferrite that links demixing free-energy curvature, first-principles elastic constants, and Rayleigh SAW observables. Free-energy curvature provides a direct stability criterion that identifies spinodally unstable composition windows. DFT-derived cubic elasticity quantifies composition-dependent stiffening trends and supplies inputs for elastodynamic prediction. Combining these elements yields experimentally actionable relationships for interpreting picosecond ultrasonics across both composition and decomposition-state variables.

Data Availability

The tabulated data, the atomistic and first-principles simulation inputs and outputs, and the scripts that generate all figures will be made publicly available at https://github.com/saleemaldajani/elastopya, with an archived DOI.

Acknowledgements

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References

[1] S. A. A. Al Dajani, B. R. Dacus, C. A. Dennett, M. G. Burke, L. Waldron, T. S. Byun, J. J. Wall, K. B.-D. Anglin, O. A. A. Al Dajani, K. J. Krakowiak, et al., Detecting thermally-induced spinodal decomposition with picosecond ultrasonics in cast austenitic stainless steels, Acta Materialia 246 (2023) 118552.

[2] S. Al Dajani, B. Dacus, C. Dennett, M. G. Burke, A. Alahmadi, K. Mukahiwa, K. Anglin, J. Wall, T. S. Byun, M. Short, Detecting thermally induced spinodal decomposition with picosecond ultrasonics in cast austenitic stainless steels, Microscopy and Microanalysis 26 (S2) (2020) 1104–1106.

[3] S. A. A. Al Dajani, Non-destructively detecting spinodal decomposition at a distance towards developing gigahertz ultrasonics for in-vessel inspection, Ph.D. thesis, Massachusetts Institute of Technology (2020).

[4] S. Pugh, Xcii. relations between the elastic moduli and the plastic properties of polycrystalline pure metals, The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science 45 (367) (1954) 823–843.

[5] Y. Chang, A correlation of the coefficients of thermal expansion of metallic solids with temperature, The Journal of Physical Chemistry 70 (4) (1966) 1310–1312.

[6] M. Fine, L. Brown, H. Marcus, Elastic constants versus melting temperature in metals, Scripta metallurgica 18 (9) (1984) 951–956.

[7] P. A. Korzhavyi, A. Ruban, J. Odqvist, J.-O. Nilsson, B. Johansson, Electronic structure and effective chemical and magnetic exchange interactions in bcc fe-cr alloys, Physical Review B 79 (5) (2009) 054202.

[8] V. I. Razumovskiy, A. V. Ruban, P. A. Korzhavyi, First-principles study of elastic properties of cr-and fe-rich fe-cr alloys, Physical Review B 84 (2) (2011) 024106.
[9] P. Hosemann, D. Frazer, M. Fratoni, A. Bolind, M. Ashby, Materials selection for nuclear applications: Challenges and opportunities, Scripta Materialia 143 (2018) 181–187.
[10] X. Du, J.-C. Zhao, Facile measurement of single-crystal elastic constants from polycrystalline samples, npj Comput. Mater. 3 (1) (2017) 17.
[11] S. A. Al Dajani, et al., Non-destructively detecting lwr structural material embrittlement using transient grating spectroscopy, in: Proc. 19th Intl. Conf. Env. Deg. Mater. Nucl. Power Systems, 2019.
[12] F. Hofmann, M. P. Short, C. A. Dennett, Transient grating spectroscopy: An ultrarapid, nondestructive materials evaluation technique, MRS Bulletin 44 (5) (2019) 392–402.
[13] T. S. Byun, Y. Yang, N. Overman, J. T. Busby, Thermal aging phenomena in cast duplex stainless steels, Jom 68 (2) (2016) 507–516.
[14] T. G. Lach, W. E. Frazier, J. Wang, A. Devaraj, T. S. Byun, Precipitation-site competition in duplex stainless steels: Cu clusters vs spinodal decomposition interfaces as nucleation sites during thermal aging, Acta Materialia 196 (2020) 456–469.
[15] S. Timoshenko, J. N. Goodier, Theory of Elasticity (1951).
[16] MIT, Constitutive Equations, 16.20 Structural Mechanics, Lecture Notes, Module 3.
[17] U. Schwingenschlögl, The interplay of structural and electronic properties in transition metal oxides, doctoralthesis, Universität Augsburg (2004).
[18] N. A. R. T. Hung, Nguyen Tuan, R. Saito, Quantum ESPRESSO Course for Solid-State Physics, CRC Press, 2022.
[19] F. Giustino, Materials Modelling using Density Functional Theory: Properties & Predictions, Oxford University Press, 2014.
[20] A. Van de Walle, P. Tiwary, M. De Jong, D. Olmsted, M. Asta, A. Dick, D. Shin, Y. Wang, L.-Q. Chen, Z.-K. Liu, Efficient stochastic generation of special quasirandom structures, Calphad 42 (2013) 13–18.
[21] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, Computational Materials Science 6 (1) (1996) 15–50. doi:10.1016/0927-0256(96)00008-0.
[22] G. Kresse, D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, Physical Review B 59 (3) (1999) 1758–1775. doi:10.1103/PhysRevB.59.1758.
[23] J. P. Perdew, K. Burke, M. Ernzerhof, Generalized gradient approximation made simple, Physical Review Letters 77 (18) (1996) 3865–3868. doi:10.1103/PhysRevLett.77.3865.
[24] S. Plimpton, Fast parallel algorithms for short-range molecular dynamics, Journal of Computational Physics 117 (1) (1995) 1–19. doi:https://doi.org/10.1006/jcph.1995.1039.
URL https://www.sciencedirect.com/science/article/pii/S002199918571039X
[25] D. R. Clarke, Materials selection guidelines for low thermal conductivity thermal barrier coatings, Surface and Coatings Technology 163 (2003) 67–74.
[26] D. Clarke, C. Levi, Materials design for the next generation thermal barrier coatings, Annual review of materials research 33 (1) (2003) 383–417.
[27] G. Bonny, R. C. Pasianot, D. Terentyev, L. Malerba, Iron chromium potential to model high-chromium ferritic alloys, Philosophical Magazine 91 (12) (2011) 1724–1746. doi:10.1080/14786435.2010.545780.
[28] D. G. Cahill, S. K. Watson, R. O. Pohl, Lower limit to the thermal conductivity of disordered crystals, Physical Review B 46 (10) (1992) 6131.
[29] M. Levesque, E. Martínez, C.-C. Fu, M. Nastar, F. Soisson, Simple concentration-dependent pair interaction model for large-scale simulations of fe-cr alloys, Physical Review B 84 (18) (2011) 184205.
[30] W. A. Dench, Adiabatic high-temperature calorimeter for the measurement of heats of alloying, Transactions of the Faraday Society 59 (1963) 1279–1292. doi:10.1039/TF9635901279.
[31] J.-O. Andersson, B. Sundman, Thermodynamic properties of the Cr–Fe system, Calphad 11 (1) (1987) 83–92. doi:10.1016/0364-5916(87)90021-6.
[32] B. Fultz, L. Anthony, J. L. Robertson, R. M. Nicklow, S. Spooner, M. Mostoller, Phonon modes and vibrational entropy of mixing in Fe-Cr, Physical Review B 52 (5) (1995) 3280–3285. doi:10.1103/PhysRevB.52.3280.
[33] M. Rae, M. Chiang, M. Islam, A. P. C. Wylie, A. Nguyen, M. Stapelberg, S. A. Al Dajani, K. Repček, T. Grabec, A. Kaplan, R. Freitas, M. P. Short, Detecting the onset and progression of spinodal decomposition using transient grating spectroscopy, arXiv:2601.06659 (2026). arXiv:2601.06659, doi:10.48550/arXiv.2601.06659.
[34] W. Xiong, P. Hedström, M. Selleby, J. Odqvist, M. Thuvander, Q. Chen, An improved thermodynamic modeling of the Fe–Cr system down to zero kelvin coupled with key experiments, Calphad 35 (3) (2011) 355–366. doi:10.1016/j.calphad.2011.05.002.
[35] C. Kittel, Introduction to Solid State Physics, 8th Edition, Wiley, Hoboken, NJ, 2005.
[36] P. Giannozzi, S. Baroni, N. Bonini, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, G. L. Chiarotti, M. Cococcioni, I. Dabo, et al., QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials, Journal of Physics: Condensed Matter 21 (39) (2009) 395502. doi:10.1088/0953-8984/21/39/395502.
[37] P. Giannozzi, O. Andreussi, T. Brumme, O. Bunau, M. B. Nardelli, M. Calandra, R. Car, C. Cavazzoni, D. Ceresoli, M. Cococcioni, et al., Advanced capabilities for materials modelling with Quantum ESPRESSO, Journal of Physics: Condensed Matter 29 (46) (2017) 465901. doi:10.1088/1361-648X/aa8f79.
[38] T. P. C. Klaver, R. Drautz, M. W. Finnis, Magnetism and thermodynamics of defect-free Fe-Cr alloys, Physical Review B 74 (9) (2006) 094435. doi:10.1103/PhysRevB.74.094435.
[39] I. M. Lifshitz, Anomalies of electron characteristics of a metal in the high pressure region, Soviet Physics JETP 11 (1960) 1130–1135.

[40] A. A. Varlamov, V. S. Egorov, A. V. Pantsulaya, Kinetic properties of metals near electronic topological transitions ($2\frac{1}{2}$-order transitions), Advances in Physics 38 (5) (1989) 469–564. doi:10.1080/00018738900101132.
[41] V. J. Minkiewicz, G. Shirane, R. Nathans, Phonon dispersion relation for iron, Physical Review 162 (3) (1967) 528–531. doi:10.1103/PhysRev.162.528.
[42] W. M. Shaw, L. D. Muhlestein, Investigation of the phonon dispersion relations of chromium by inelastic neutron scattering, Physical Review B 4 (3) (1971) 969–973. doi:10.1103/PhysRevB.4.969.
[43] L. Mauger, M. S. Lucas, J. A. Muñoz, S. J. Tracy, M. Kresch, Y. Xiao, P. Chow, B. Fultz, Nonharmonic phonons in α-iron at high temperatures, Physical Review B 90 (2014) 064303. doi:10.1103/PhysRevB.90.064303.
[44] E. Fawcett, Spin-density-wave antiferromagnetism in chromium, Reviews of Modern Physics 60 (1) (1988) 209–283. doi:10.1103/RevModPhys.60.209.
[45] R. Hafner, D. Spišák, R. Lorenz, J. Hafner, Does density-functional theory predict a spin-density-wave ground state for Cr?, Journal of Physics: Condensed Matter 13 (2001) L239–L247.
[46] S. Cottenier, B. De Vries, J. Meersschaut, M. Rots, What density-functional theory can tell us about the spin-density wave in Cr, Journal of Physics: Condensed Matter 14 (2002) 3275–3283.
[47] V. L. Moruzzi, J. F. Janak, A. R. Williams, Calculated Electronic Properties of Metals, Pergamon, New York, 1978.
[48] P. Olsson, I. A. Abrikosov, L. Vitos, J. Wallenius, Ab initio formation energies of Fe–Cr alloys, Journal of Nuclear Materials 321 (1) (2003) 84–90. doi:10.1016/S0022-3115(03)00207-1.
[49] D. G. Pettifor, Theoretical predictions of structure and related properties of intermetallics, Materials Science and Technology 8 (4) (1992) 345–349. doi:10.1179/mst.1992.8.4.345.

Supplementary Material

~~The following material retains~~

S1. Thermal transport estimates

Thermal properties in Fig. S7 are estimated with the Debye model, and a lower limit for the thermal conductivity is obtained at high temperature [25,26]:

$$\kappa = \frac{1}{3} C_v v_m \Lambda \tag{7}$$

$$\kappa_{min} \to k_B v_m \Lambda \to k_B v_m \Lambda_{min} = A k_B \left(\frac{N_A m}{M}\right)^{\frac{2}{3}} \rho^{\frac{1}{6}} E^{\frac{1}{2}} \tag{8}$$

$$\Theta_D = \frac{2\hbar v_m}{a_0 k_B} \to B_D \frac{\hbar}{k_B} \left(\frac{N_A m}{M}\right)^{\frac{1}{3}} \frac{E^{\frac{1}{2}}}{\rho^{\frac{1}{6}}} \tag{9}$$

where κ is thermal conductivity, Θ_D is the Debye temperature, C_v is the heat capacity at constant volume, v_m is the average acoustic velocity, Λ is the phonon mean free path, m is the number of atoms, M is the mass, ρ is the density, and E is Young's modulus. The constants A and B_D arise from approximating v_m as proportional to $\sqrt{E/\rho}$: Over a wide range of Poisson's ratios, $v_m \approx A\sqrt{E/\rho}$ with $A = 0.87 \pm 0.02$, and the same approximation gives $B_D = A(6\pi^2)^{1/3} = 3.39$ (Clarke [25], Eqs. 8–10, p. 70); we use these values. Evaluating the Debye average of the transverse and longitudinal velocities directly instead gives $v_m = 0.70\sqrt{E/\rho}$ for $\nu = 0.25$–0.30. With this value and the measured Young's modulus (211 GPa), the Debye temperature of bcc Fe is 471 K, matching the calorimetric 470 K [35], whereas $B_D = 3.39$ gives 589 K. The absolute κ_{min} and D therefore carry an uncertainty of about 20% from the choice of A. Composition and decomposition trends do not depend on A. A temperature-dependent alternative is the Cahill–Watson–Pohl model, $\kappa_{min}(T) = (\frac{\pi}{6})^{\frac{1}{3}} k_B n^{\frac{2}{3}} \sum_i v_i (\frac{T}{\Theta_i})^2 \int_0^{\Theta_i/T} \frac{x^3 e^x}{(e^x-1)^2}\, dx$ with $\Theta_i = v_i(\hbar/k_B)(6\pi^2 n)^{\frac{1}{3}}$ [28].

S2. Additional figures

The following figures are retained from the original draft ~~figures~~ for completeness and traceability~~. All figure numbering in this supplementary section is independent of the main manuscript~~; Sections S3–S9 contain the material added in this revision.

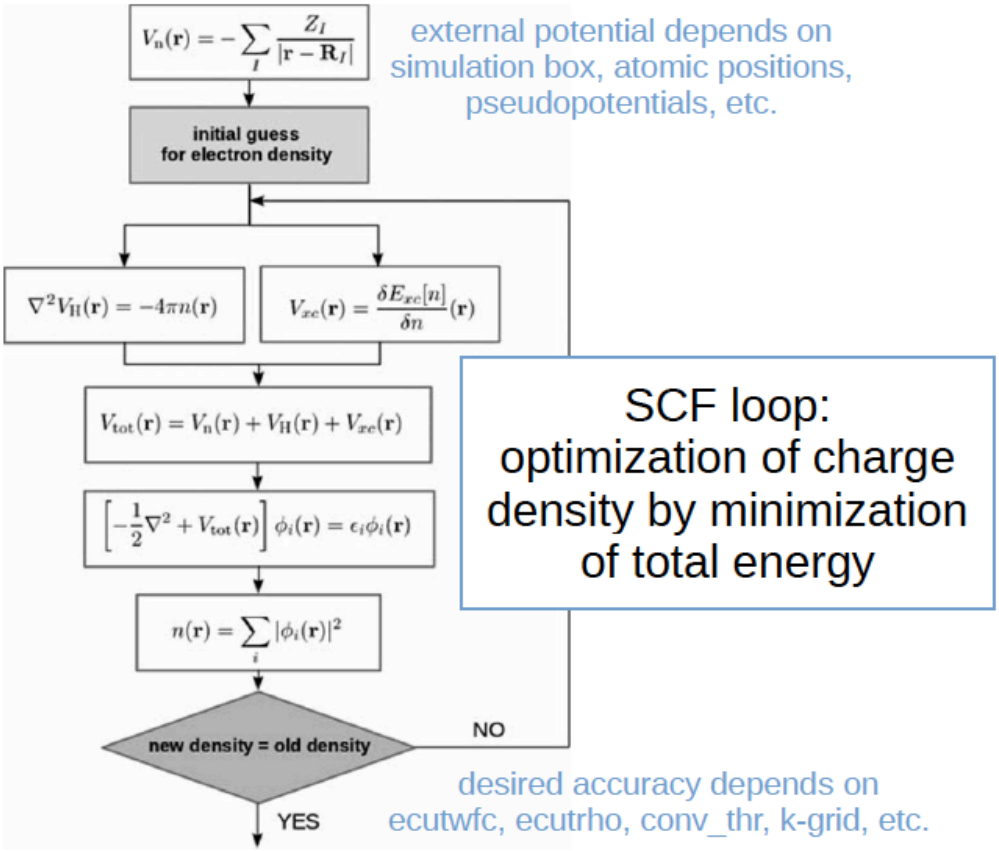


Figure S1: Self-consistent field (SCF) loop schematic illustrating self-consistent electronic-structure workflow [19].

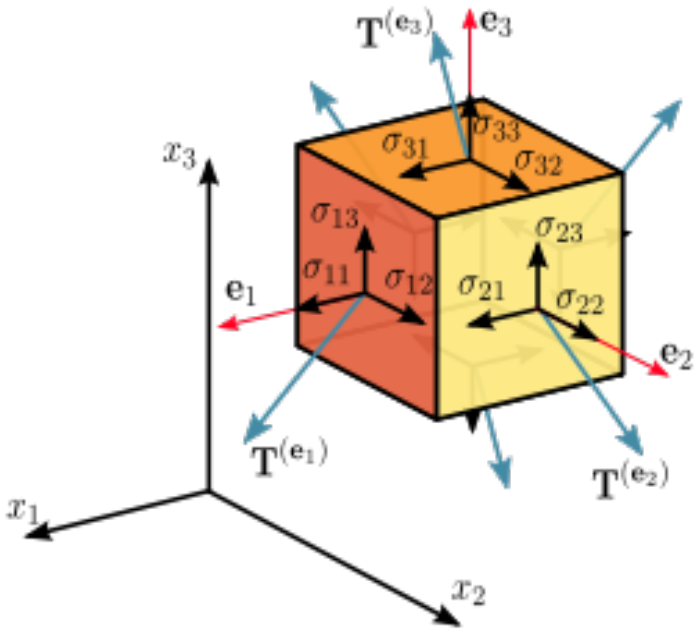
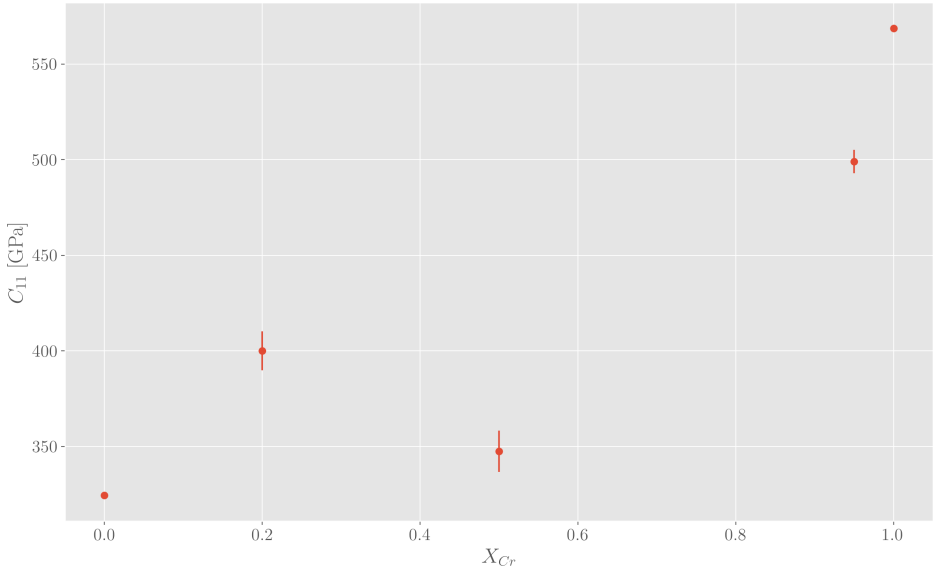
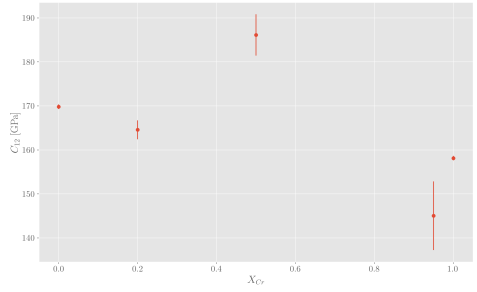


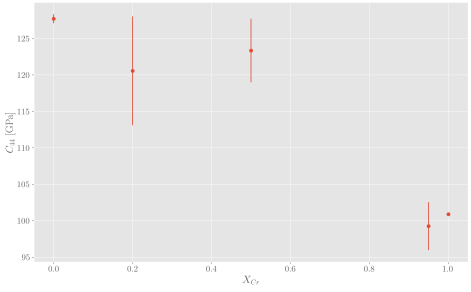
Figure S2: Stress tensor visualization [3].



(a) Normal constant, C_{11}, illustrating the longitudinal stiffness of bcc binary FeCr alloy as a function of chromium composition. This stiffness parameter is crucial for understanding the acoustic velocity and mechanical properties of the material.

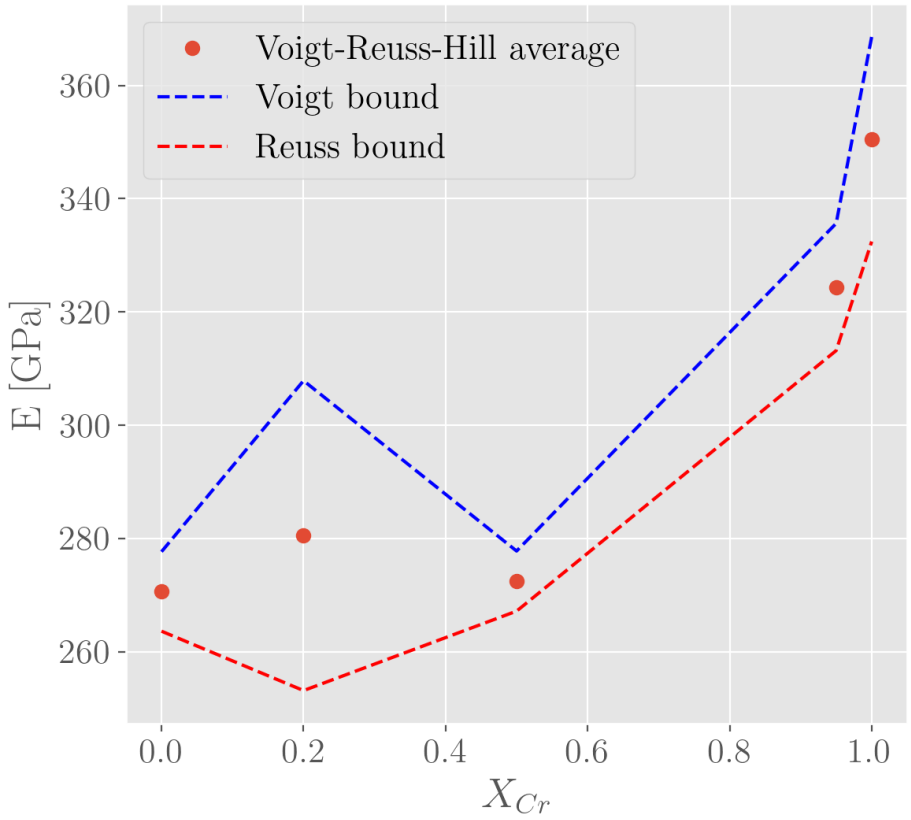


(b) The interplanar elastic constant C_{12}, showing the coupling between axial and lateral strains, indicative of the alloy's resistance to shear along crystallographic planes.

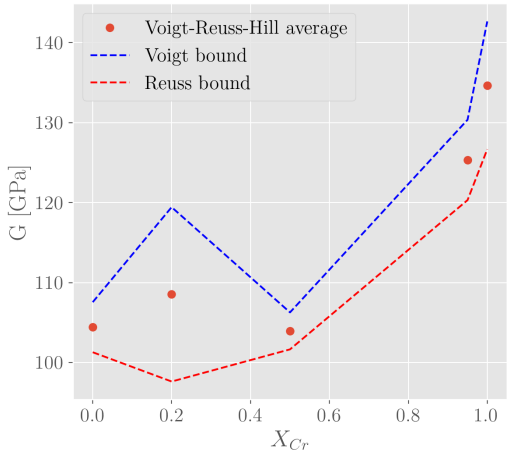


(c) Shear constant, C_{44}, representing shear stiffness against deformation without volume change, critical for assessing material's resistance to applied shear stress.

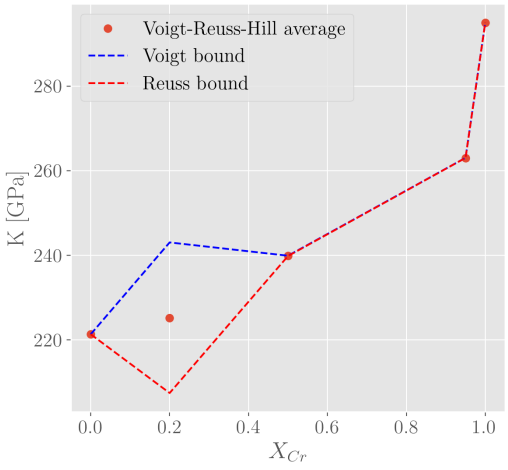
Figure S3: Elastic constants of homogeneous bcc ~~binary~~ FeCr from the present DFT calculations (40-atom SQS) as a function of Cr composition~~indicate spinodal decomposition-induced stiffening towards embrittlement~~. ~~The trends and values are consistent with theoretical and experimental results reported in literature.~~ (a) Shows the longitudinal stiffness parameter C_{11}, important for evaluating the material's response to compressive and tensile stresses. (b) Highlights C_{12}, which is essential for understanding the coupling between different types of strain. (c) Details C_{44}, crucial for evaluating the shear deformation behavior.



(a) Elastic (Young's) Modulus, E_{VRH}, indicating the material's stiffness and its ability to resist deformation under tensile stress.

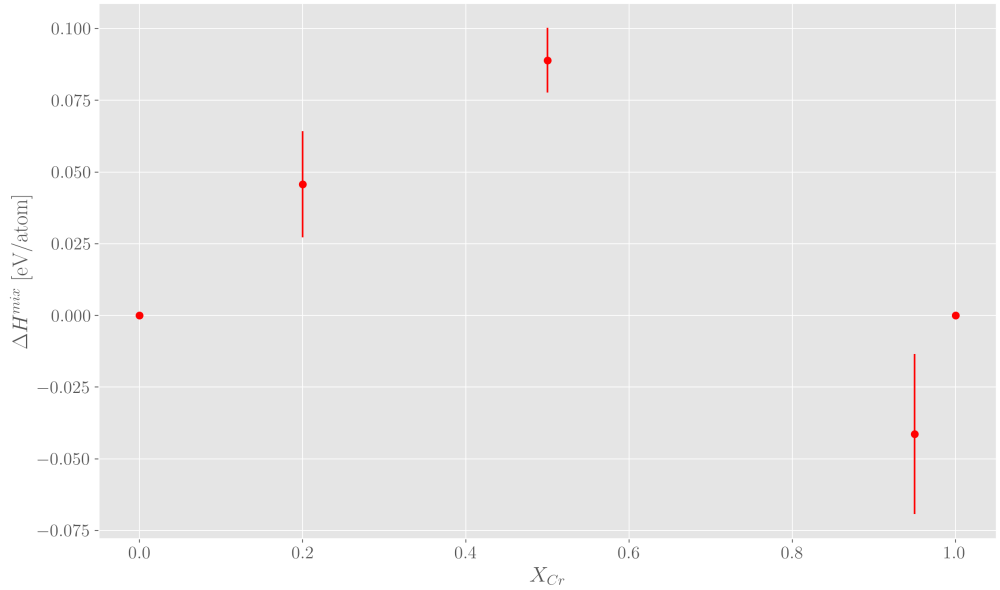


(b) Shear modulus, G_{VRH}, reflecting the material's ability to withstand shear deformation.

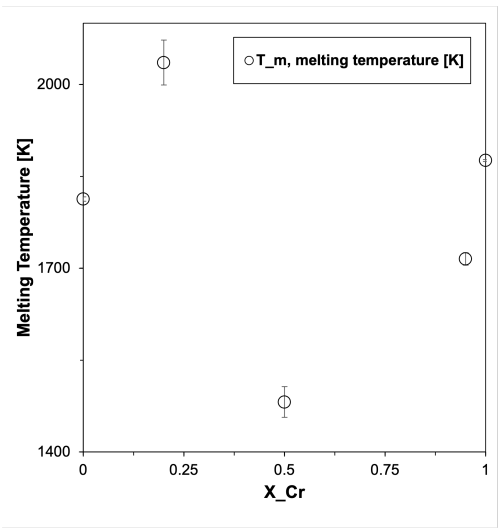


(c) Bulk modulus, K_{VRH}, which measures the material's resistance to uniform compression.

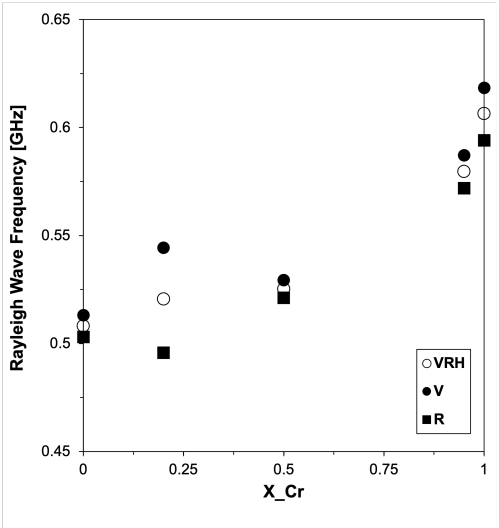
Figure S4: ~~Elastic~~ VRH elastic moduli of homogeneous bcc ~~binary~~ FeCr from the present DFT calculations as a function of Cr composition~~indicate spinodal decomposition-induced stiffening towards embrittlement. The trends and values are consistent with theoretical and experimental results reported in literature.~~ (a) Elastic (Young's) Modulus demonstrates the material's overall stiffness. (b) Shear modulus is a measure of the alloy's rigidity against shear forces. (c) Bulk modulus assesses the material's compressibility under uniform pressure.



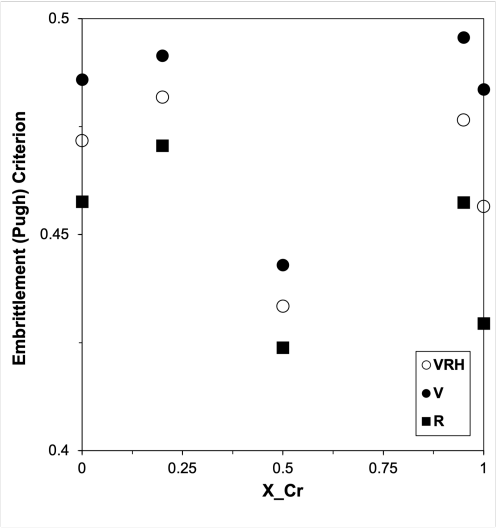
(a) Enthalpy of mixing, showing the thermodynamic favorability of phase separation in bcc δ-ferrite in cast austenitic stainless steels through spinodal decomposition.



(b) T_M, melting temperature, indicating how the thermal stability of the material changes with Cr composition.

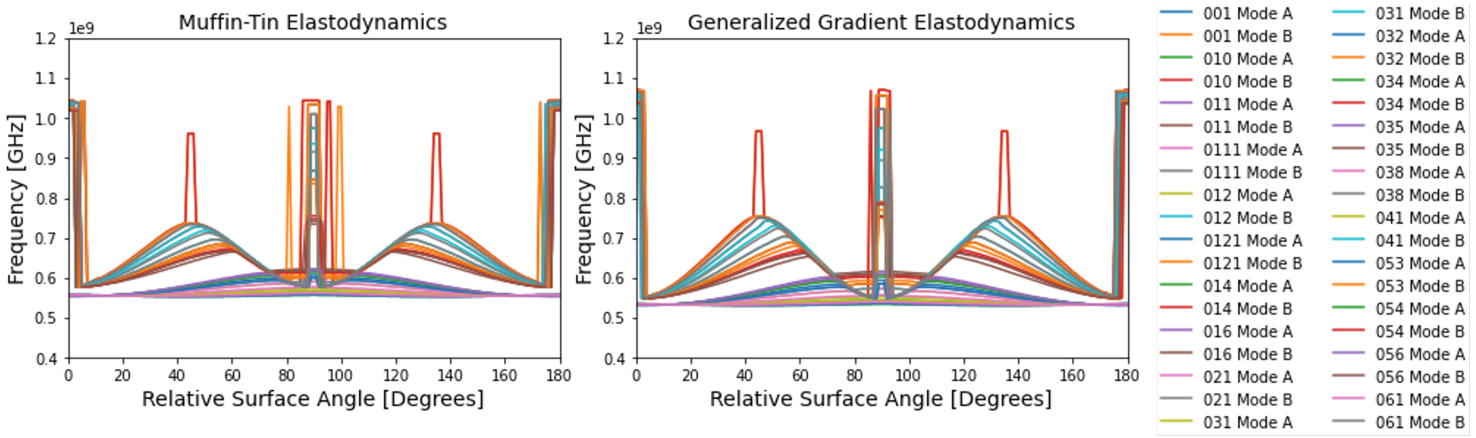


(c) Rayleigh wave frequencies, illustrating the impact of spinodal decomposition on the material's surface acoustic properties.

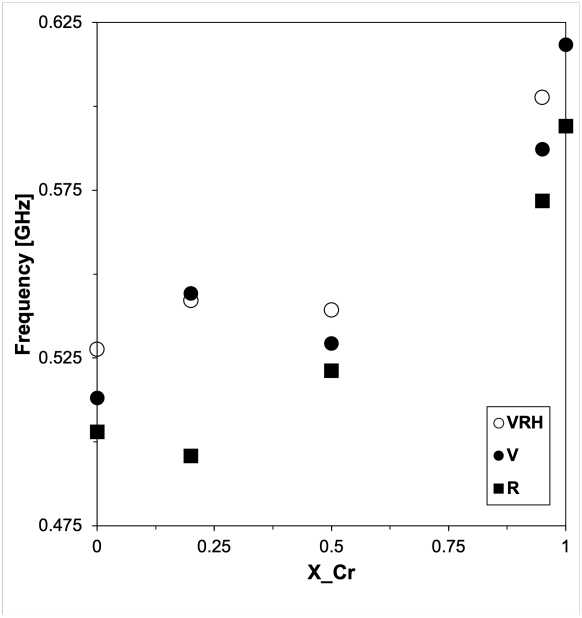


(d) Embrittlement (Pugh) Criterion, calculated to evaluate the embrittlement risk associated with varying Cr compositions.

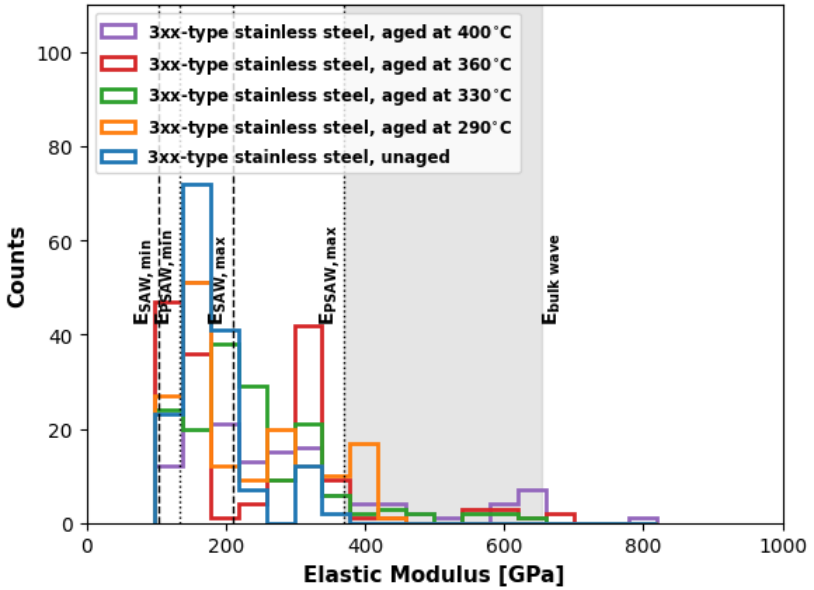
Figure S5: (a) The enthalpy of mixing demonstrates the thermodynamic drive towards phase separation indicative of spinodal decomposition, affecting the material's mechanical integrity. (b) Variations in the melting temperature as a function of Cr composition offer insights into the thermal stability and potential phase transformations within the alloy. (c) Changes in Rayleigh wave frequencies across compositions highlight alterations in surface acoustic properties, correlating with mechanical behavior. (d) The Embrittlement (Pugh) Criterion allows for assessing the risk of material embrittlement due to changes in composition, crucial for ensuring material reliability in critical applications.



(a) Predicted elastic frequencies showcasing the emergence of new modes within the elastodynamically-unallowed regions, signifying alterations in the material's mechanical properties due to spinodal decomposition.~~These frequencies match closely with experimentally observed values in aged stainless steel specimens, providing a quantum mechanical insight into the microstructural evolution impacting material performance.~~

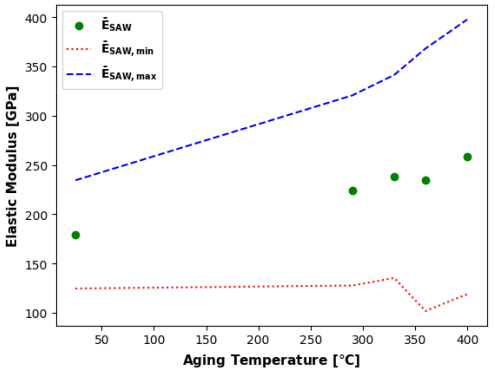


(b) Calculated acoustic frequencies across different Cr compositions, illustrating the material's acoustic response variability with composition. This graph provides a quantitative measure of the material's mechanical behavior and its dependence on the microstructural characteristics induced by varying chromium content.

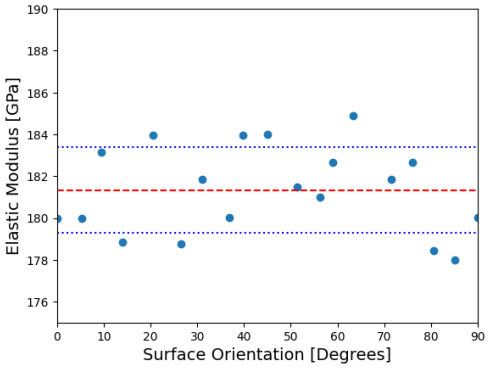


(c) Measured elastic moduli distribution from experimental data, highlighting the range and variability of material properties within stainless steel samples subjected to different thermal aging processes. The distribution underscores the significant impact of microstructural changes, such as spinodal decomposition, on the material's overall stiffness and mechanical integrity.

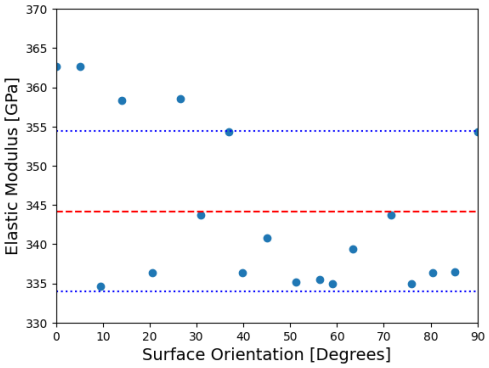
Figure S6: (a) Quantum elastodynamic frequencies obtained from density functional theory calculations reveal the complex interplay between microstructural evolution and mechanical properties in stainless steels undergoing spinodal decomposition. These findings are corroborated by (b) calculated acoustic frequencies that vary with Cr composition, providing a link between composition, microstructure, and acoustic properties. (c) The distribution of measured elastic moduli further validates the theoretical predictions, illustrating the broad impact of aging and compositional changes on the material's mechanical performance.



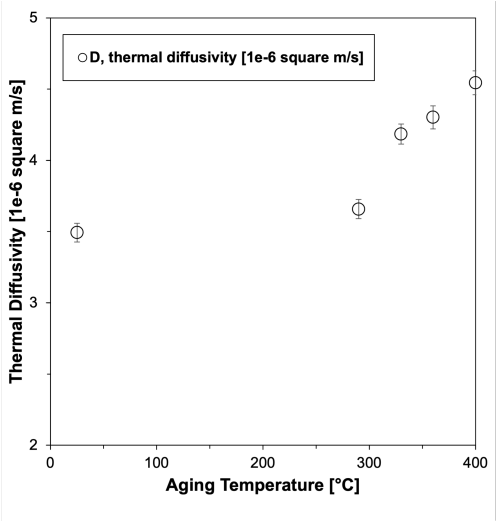
(a) Experimental elastic moduli of aged steels, displaying how aging temperature influences the material's stiffness. This graph illustrates the correlation between thermal treatment and the increase in elastic moduli, indicative of the material's response to microstructural changes over time.



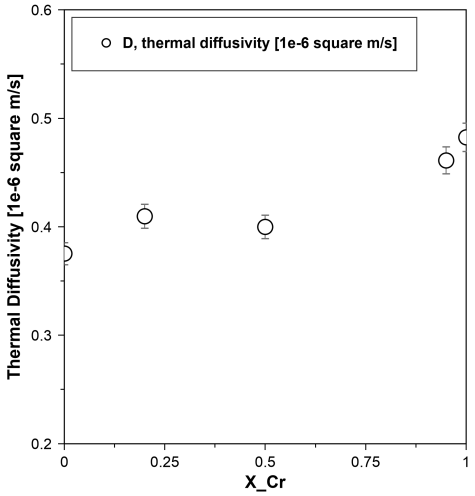
(b) Theoretical elastic moduli of pristine steels, offering a baseline comparison for understanding how spinodal decomposition affects material properties before and after aging. This comparison elucidates the stiffening effect of microstructural evolution on the alloy's mechanical behavior.



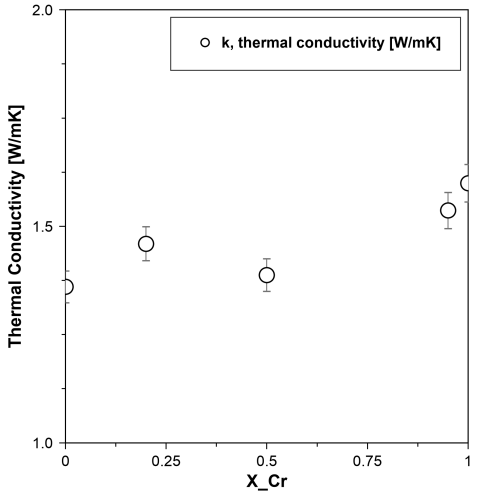
(c) Theoretical elastic moduli of spinodally decomposed steels, demonstrating the pronounced impact of microstructural evolution on the alloy's stiffness. This direct comparison to pristine conditions highlights the significant role of chromium composition and thermal aging in material performance.



(d) $D^{exp}_{thermal}$, experimental thermal diffusivity as measured in aged stainless steel, illustrating the thermal transport characteristics and their variation with temperature. This metric provides insights into the thermal stability and heat dissipation capabilities of the material post-aging.



(e) $D^{theoretical}_{thermal}$, theoretical thermal diffusivity indicating the expected thermal behavior based on composition and microstructural considerations. This predictive analysis aids in understanding the thermal management potential of the alloy across a range of compositions.



(f) $\kappa^{theoretical}_{thermal}$, thermal conductivity, showcasing the theoretical prediction of how efficiently the material conducts heat. This property is crucial for applications where thermal management is key, and it is influenced by the alloy's microstructure and composition.

Figure S7: Comprehensive analysis combining experimental and theoretical insights into the thermal and mechanical properties of stainless steels subjected to thermal aging. (a) Experimental elastic moduli trends with aging temperature highlight the stiffening effect of microstructural evolution. (b) and (c) Theoretical predictions for pristine and spinodally decomposed steels provide a comparative framework for assessing the impact of spinodal decomposition on material properties. (d) Experimental thermal diffusivity measurements, alongside (e) theoretical predictions for thermal diffusivity and (f) thermal conductivity, offer a holistic view of the material's thermal and mechanical behavior post-aging, emphasizing the interconnectedness of thermal properties, mechanical performance, and microstructural characteristics.

S3. First-principles calculations: methods and overview

Methods. An independent set of first-principles calculations was performed with Quantum ESPRESSO 7.5 [36,37] on the MIT Engaging cluster, using PBE [23] and PAW potentials from pslibrary (Fe 3s3p3d4s, Cr 3s3p3d4s), cutoffs of 60/480 Ry (wavefunctions/density), and Marzari–Vanderbilt smearing of 0.02 Ry. The alloys are 40-atom SQS ($2 \times 2 \times 5$ conventional bcc cells; pair correlations optimised over three neighbour shells) at $x_{Cr} = 0.20$, 0.35, 0.50, 0.65 and 0.95, sampled with a $4 \times 4 \times 2$ k-mesh. Starting moments are ferromagnetic, and each cell is fully relaxed (box lengths and ions). Elastic constants follow from the stress response to strains $\varepsilon_{xx} = \pm 0.5\%$ and $\gamma_{yz} = \pm 0.5\%$ with ions relaxed at each strain (central differences). Pure Fe (FM) and Cr (NM and commensurate AFM) use 2-atom cells with 16^3 meshes. Convergence for bcc Fe: C_{ij} change by ≤ 2 GPa between 60 and 80 Ry and are converged by a 12^3 mesh. Electronic structure: atom- and spin-projected DOS from a non-self-consistent $6 \times 6 \times 3$ run with the optimised tetrahedron method; Fermi-surface sections on the $k_z = 0$ plane; band structures along Γ–H–N–Γ–P–H. Phonons: density-functional perturbation theory on a 6^3 q-grid, Fourier-interpolated along the same path.

Overview. The additional first-principles material is organised by topic: elastic constants at the PBE-relaxed and the room-temperature experimental lattice parameter (Section S4); electronic structure, including the unfolded Fermi surfaces and band structures of the Fe-rich alloys (S5); the electronic topological transition that links them to the Fe-rich elastic anomaly (S6); lattice dynamics (S7); a validation of every computed quantity against published data and experiment, including the thermal-transport estimates of Section S1 (S8); and the proposed mechanism figures (S9). Each section gives its text first and its figures after.

To be updated: still running: the PBE-relaxed 10 pct Cr unfolded Fermi surface and the relaxed 5 and 10 pct Cr unfolded bands; Fe-80Cr elastic constants at the PBE-relaxed lattice parameter; the Cr nesting function. Planned: SQS phonon DOS with the vibrational entropy of mixing.

S4. Elastic constants

Figure S8 compares the Fe-rich elastic constants with EMTO-CPA and MD. Figure S9 repeats the QE calculations at the room-temperature experimental lattice parameter (Vegard between Fe 2.8665 and Cr 2.8846 Å; cubic cell fixed, ions relaxed), the convention of the EMTO-CPA reference data [8]. PBE relaxes all compositions to about 11.4 Å^3/atom, about 4% below experiment; the bulk modulus is the property most sensitive to this (Section S8).

At the experimental lattice parameter f_{SAW} rises from 446 MHz for Fe (EMTO-CPA: 449 MHz) to 485 at 20%, flattens at 497, 500 and 499 MHz between 30 and 40% Cr (EMTO-CPA: 503, 488, 490 MHz), and reaches 513 at 50% (513), 542 at 80% and 580 MHz for AFM Cr (602). The bulk modulus has a local minimum at 40% Cr (163 GPa) where the Cr moments collapse, and the shear modulus flattens there. The experimental lattice parameter thus reproduces the magnitude of f_{SAW} and a magnetic anomaly near Fe–36Cr, but not the Fe-rich maximum near 20% Cr of the EMTO-CPA curve, so it gives a plateau rather than a minimum. At the PBE-relaxed volume f_{SAW} rises steadily from 504 to 591 MHz. The bulk-modulus drops at 80% and 100% Cr reflect antiferromagnetic (Cr) and ferrimagnetic (Fe–80Cr, Cr antiparallel to Fe, net 0.31 μ_B/atom) order, which collapses C_{12} as in pure Cr.

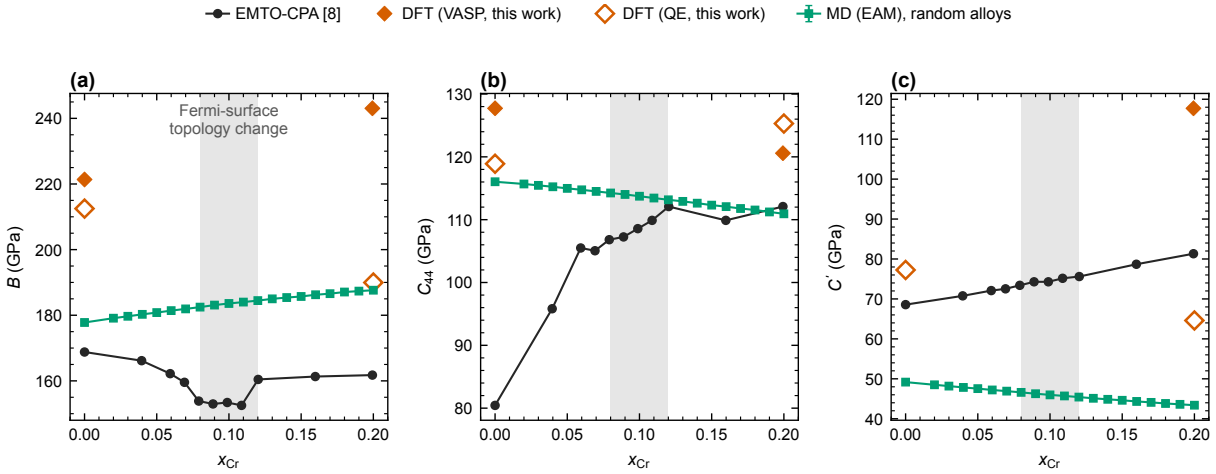


Figure S8: Fe-rich elastic constants ($x_{Cr} \leq 0.2$): bulk modulus B, C_{44} and C' from EMTO-CPA [8], MD (EAM) random alloys on a 1% composition grid, and the present VASP and Quantum ESPRESSO calculations. Validation against measured bulk moduli (compiled in Ref. [7]): at the relaxed PBE volume both DFT codes overestimate B (VASP: Fe 221 vs 168 GPa, +32%; QE SQS at 95% Cr: +31% against the paramagnetic value), the usual consequence of the too-small PBE volume of Fe–Cr; at the room-temperature experimental lattice parameter the QE value at 95% Cr is within 7%. The shaded band (0.08–0.12) marks the composition range of the Fermi-surface topology change reported in Ref. [8]; the classical potential has no electronic structure and does not reproduce it.

Spinodal window from the QE enthalpies. The complete QE enthalpies of mixing (Fig. 1a) can be fitted with Redlich–Kister polynomials, $\Delta H_{mix} = x(1-x)\sum_k L_k(1-2x)^k$, and combined with the ideal configurational entropy to give the demixing curvature (Fig. S10). With four terms (RMS 2.2 and 4.3 meV/atom at the PBE-relaxed and experimental lattice parameters) the alloy is unstable at 748 K for $0.23 \leq x_{Cr} \leq 0.99$ at both lattice parameters. With three terms (RMS 4.0 and 6.2 meV/atom) the window is 0.34–0.96 and 0.37–0.97, which places Fe–36Cr on its lower edge. The lower edge therefore depends on the fit and is not determined by the present data to better than about 0.1. The upper edge lies far above the CALPHAD value of 0.71 [34] because these are 0 K ferromagnetic energies, which omit the paramagnetism of Cr-rich alloys at the aging temperature. We therefore keep the CALPHAD window (0.30–0.71 at 748 K) as the reference in the main text and use the QE enthalpies only for the shape of ΔG_{demix} in Fig. 1a.

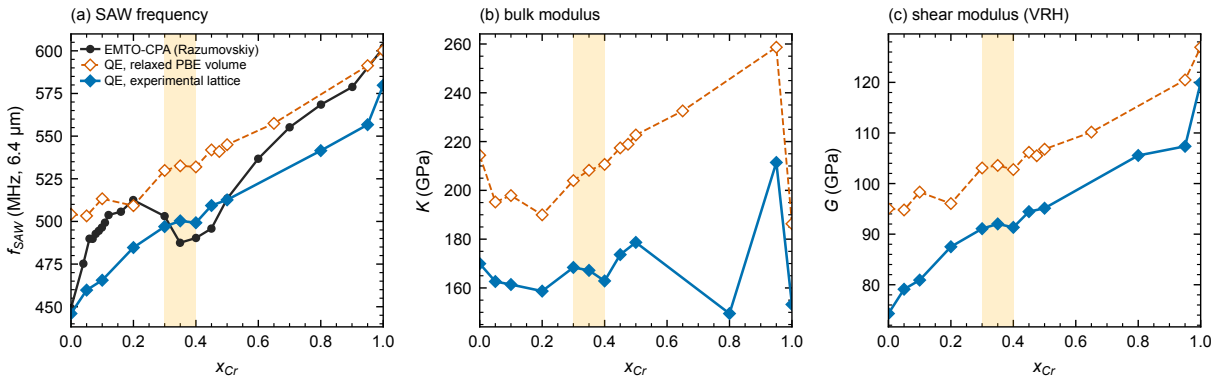


Figure S9: QE 40-atom SQS and pure elements at the PBE-relaxed (open, dashed) and the room-temperature experimental (filled) lattice parameter, against EMTO-CPA [8]: (a) f_{SAW} at $\lambda = 6.4\,\mu$m, (b) bulk modulus, (c) VRH shear modulus. The shaded band marks 30–40% Cr. Pure Cr is AFM; Fe–80Cr at the experimental lattice parameter converged to a ferrimagnetic state.

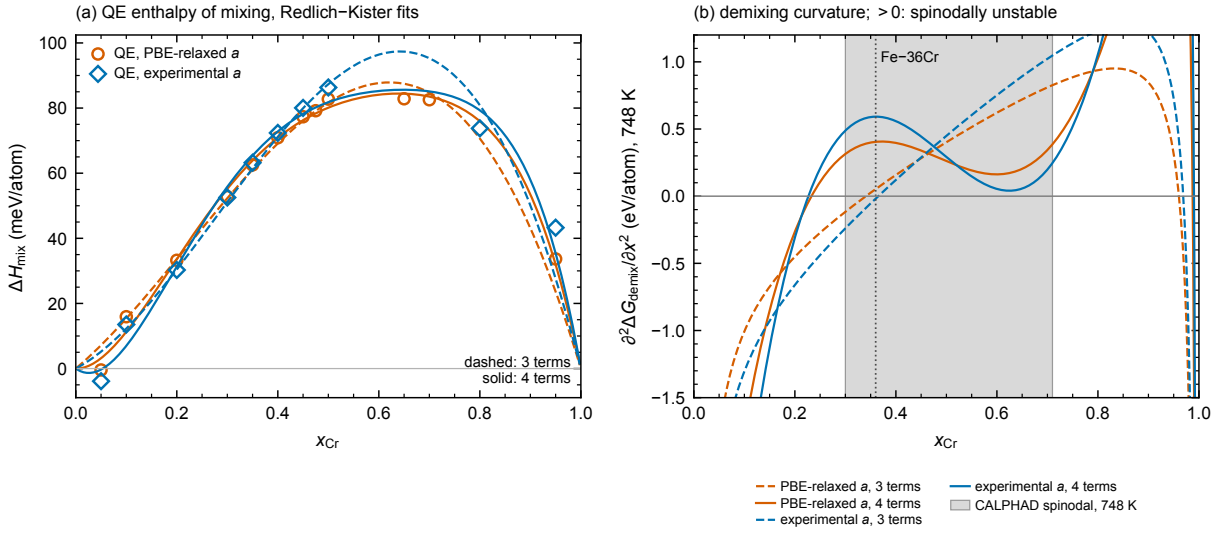


Figure S10: Spinodal window implied by the QE enthalpies of mixing. (a) QE ΔH_{mix} of the 40-atom SQS at the PBE-relaxed (circles) and experimental (diamonds) lattice parameter, with three-term (dashed) and four-term (solid) Redlich–Kister fits. (b) Demixing curvature $\partial^2 \Delta G_{demix}/\partial x^2$ at 748 K from each fit with the ideal configurational entropy; positive values are spinodally unstable. Grey band: CALPHAD chemical spinodal at 748 K [34]; dotted line: Fe–36Cr. The lower edge of the QE window moves from 0.23 to 0.34–0.37 with the fit order; the upper edge (0.96–0.99) reflects the 0 K ferromagnetic reference. Script: scripts/make_qe_spinodal_sm.py.

S5. Electronic structure

Fermi surfaces and band structures of the pure metals (Figs. S11–S14) set the reference for the alloys. The unfolded Fermi surfaces and band structures of the Fe-rich SQS (Figs. S15 and S16) follow how Cr changes the spin-up Fermi surface; the projected densities of states of the SQS (Fig. S18) show the composition dependence of the Fe and Cr states and of the Fermi-level DOS.

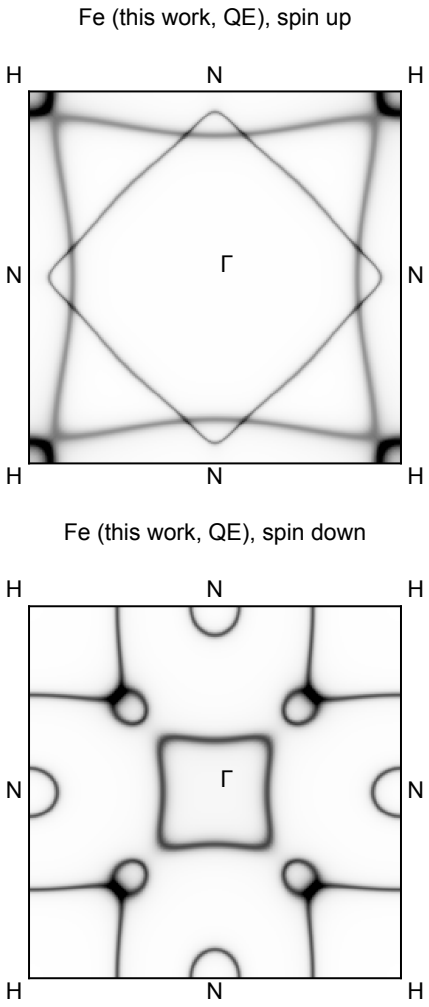


Figure S11: Fermi surface of ferromagnetic bcc Fe in the (001) plane through Γ (square with H at the corners), spin up and spin down, shown as a broadened spectral function in the layout of Korzhavyi et al. [7] (their Fig. 6).

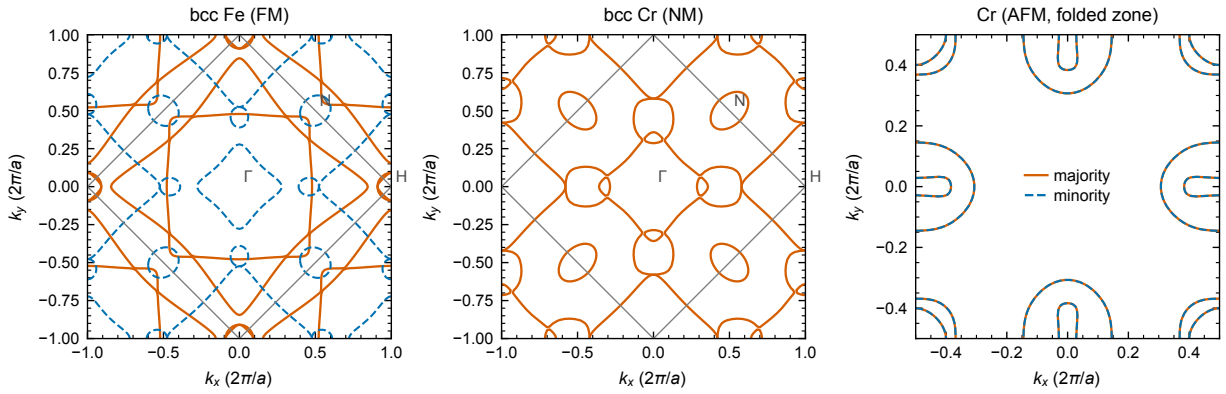


Figure S12: Fermi-surface sections ($k_z = 0$) of bcc Fe (FM; solid, majority; dashed, minority), bcc Cr (NM) and commensurate AFM Cr (folded simple-cubic zone; both spin channels coincide). The nested electron and hole sheets of Cr are responsible for its incommensurate spin-density wave.

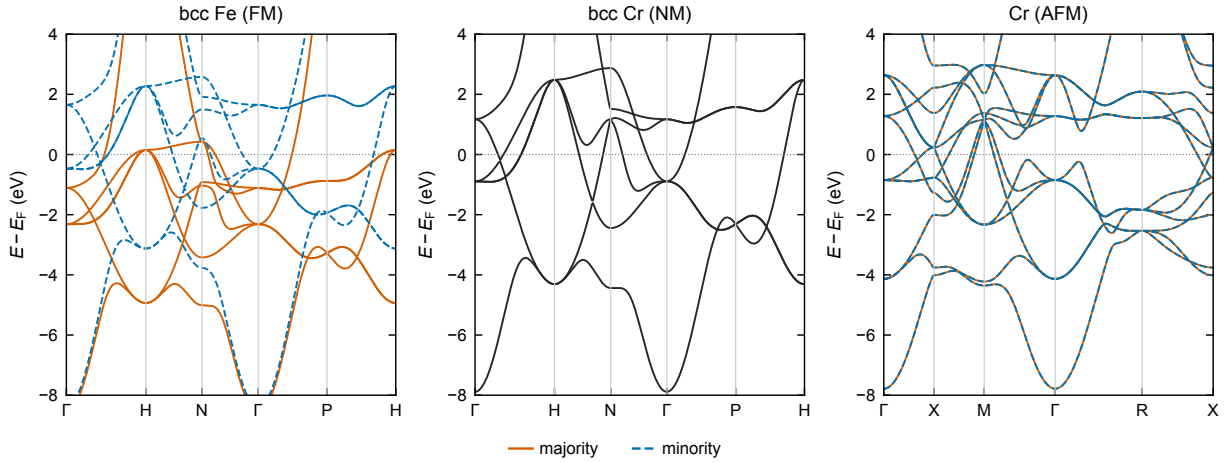


Figure S13: Electronic band structures of bcc Fe (FM), bcc Cr (NM) and AFM Cr (simple-cubic path), Quantum ESPRESSO. Validation: along Γ–H the Fe bands within ±2.7 eV of E_F agree with the EMTO-CPA spectrum of pure Fe (Korzhavyi et al. [7], Fig. 7) to an RMS of 0.22 eV (spin up) and 0.17 eV (spin down); the band crossings at E_F and the H-point level 0.15 eV above E_F in the majority channel coincide.

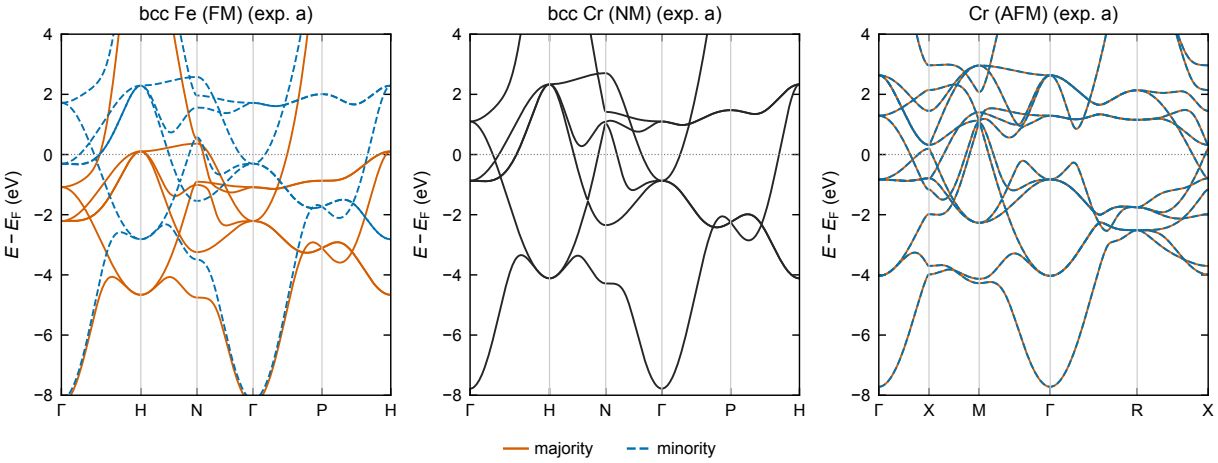


Figure S14: Electronic band structures of bcc Fe (FM), bcc Cr (NM) and AFM Cr at the room-temperature experimental lattice parameters (Fe 2.8665 Å, Cr 2.8846 Å), for comparison with Fig. S13 at the relaxed PBE lattice parameters. Validation: the Fe Γ–H bands agree with EMTO-CPA [7] to an RMS of 0.18 eV (spin up) and 0.14 eV (spin down), slightly better than at the relaxed lattice parameter.

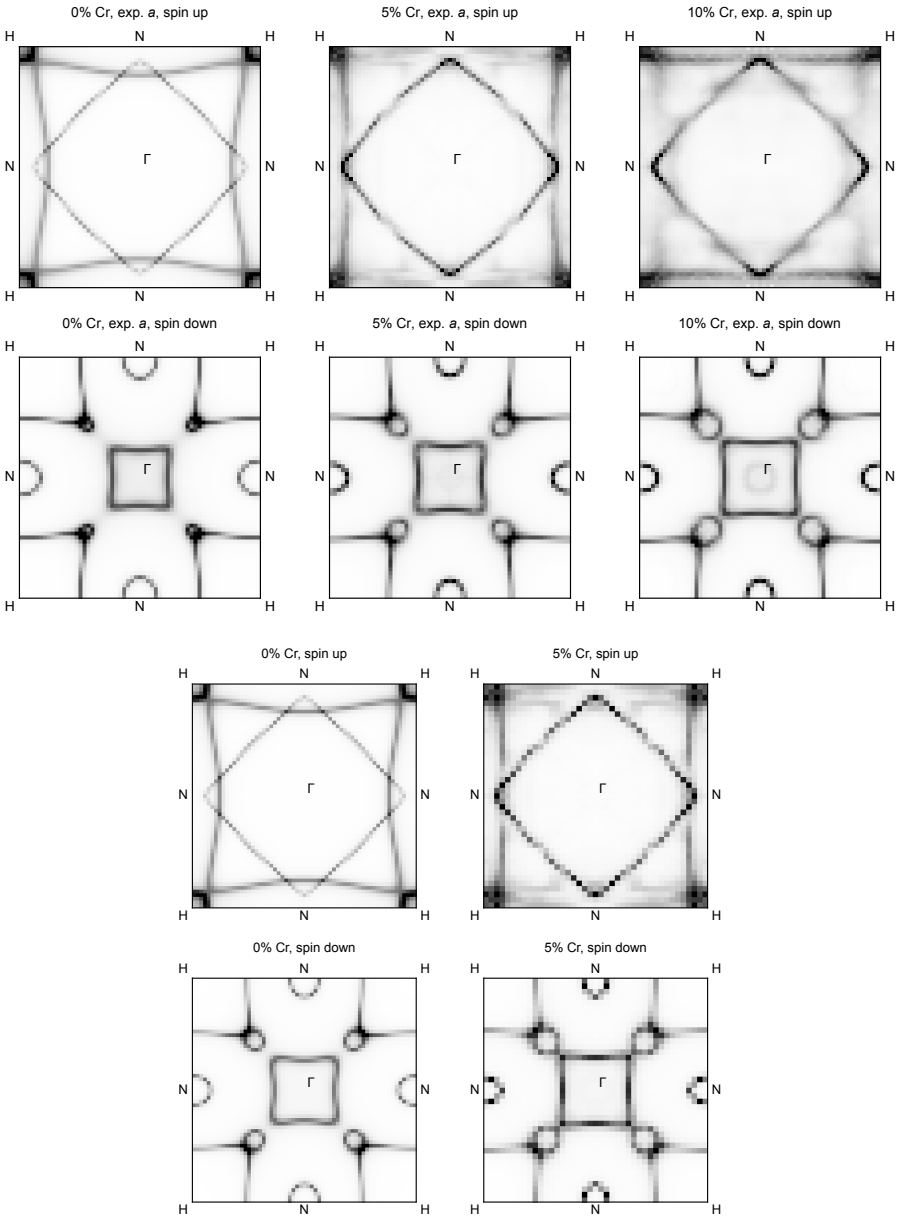


Figure S15: Unfolded Fermi surfaces (Bloch spectral function at E_F, $k_z = 0$; spin up top, spin down bottom) of the 40-atom SQS, in the layout of Korzhavyi et al. [7] (their Fig. 6). Top: 0, 5 and 10% Cr at the experimental lattice parameter. Bottom: 0 and 5% Cr at the PBE-relaxed lattice parameter. The supercell bands were unfolded onto the bcc zone with BandUPpy (band step with ParO diagonalisation), on a 61×61 grid except for the relaxed 5% Cr panel (41×41, hence coarser pixels). Pure Fe reproduces the Fermi surface of the primitive cell (Fig. S11) at both lattice parameters. With Cr the spin-up sheet changes topology: the arms that connect the H pockets through N lose their weight and the Γ-centred sheet reaches N. The change is already clear at 5% Cr at both lattice parameters, and 10% Cr looks the same with more disorder broadening, while the spin-down sheets barely change. This is the spin-up topology change that Ref. [7] places between 5 and 10% Cr, the electronic origin of the Fe-rich elastic anomaly (Fig. S8); in our SQS it sets in slightly earlier, consistent with the critical level of Fig. S19a reaching E_F at 5% Cr. To be updated: the relaxed 10 pct Cr panel will be added.

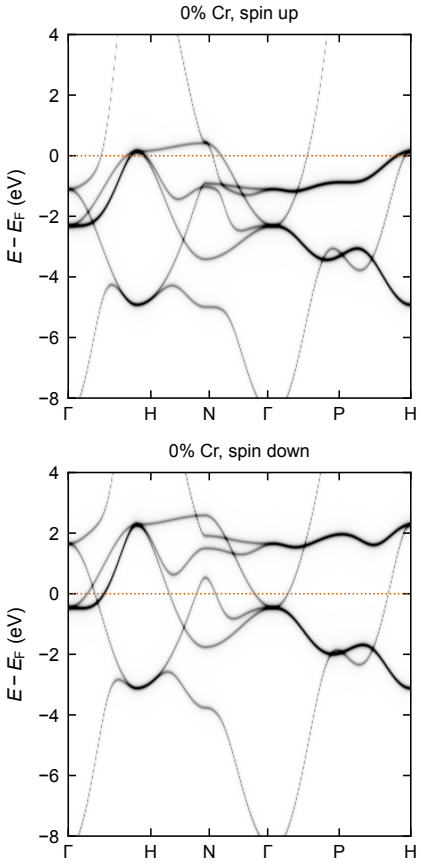


Figure S16: Unfolded (effective) band structure of the 40-atom SQS of pure Fe along Γ–H–N–Γ–P–H, spin up (top) and down (bottom), Lorentzian-broadened by 0.06 eV. It reproduces the primitive-cell bands (Fig. S13), including the majority H level 0.15 eV above E_F, which validates the unfolding before it is applied to the alloys (Fig. S17).

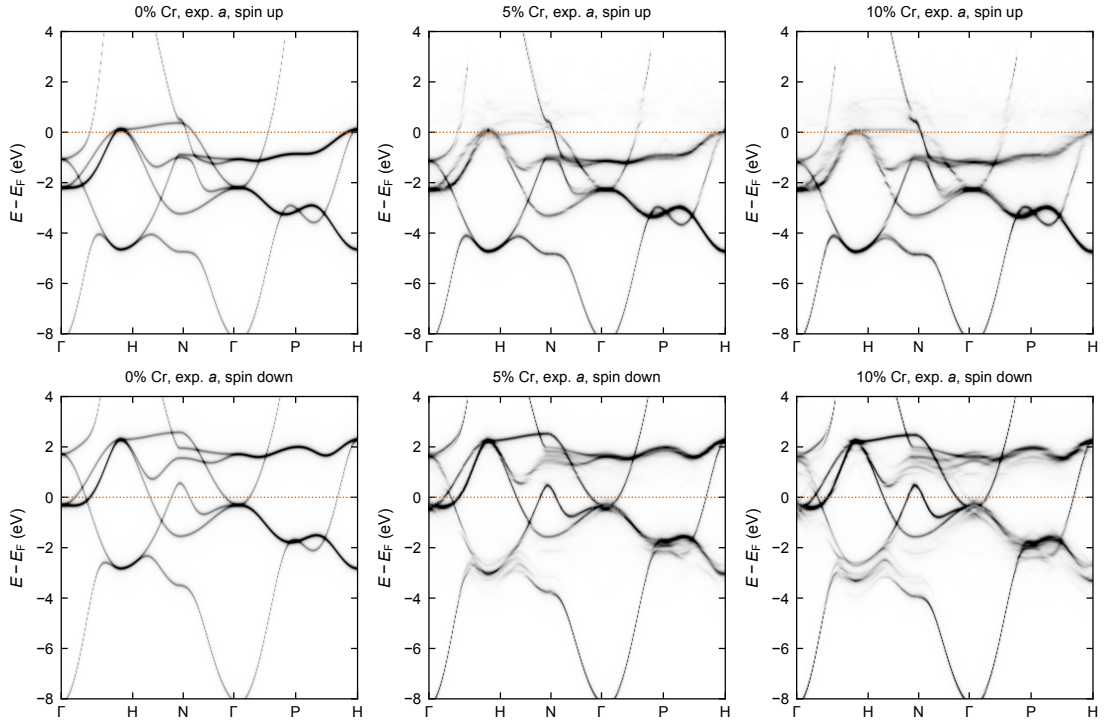


Figure S17: Unfolded band structures of the 40-atom SQS at 0, 5 and 10% Cr at the experimental lattice parameter, spin up (top) and down (bottom), as in Fig. S16. The 0% Cr cell reproduces the primitive-cell bands of Fig. S14. Cr broadens the bands (disorder) and moves the majority level at H from 0.104 eV above E_F (0%) to 0.005 eV above (5%) and 0.009 eV below (10%), with a disorder spread of about 0.1 eV; these are the points of Fig. S19a. To be updated: the PBE-relaxed 5 and 10 pct Cr alloys will be added.

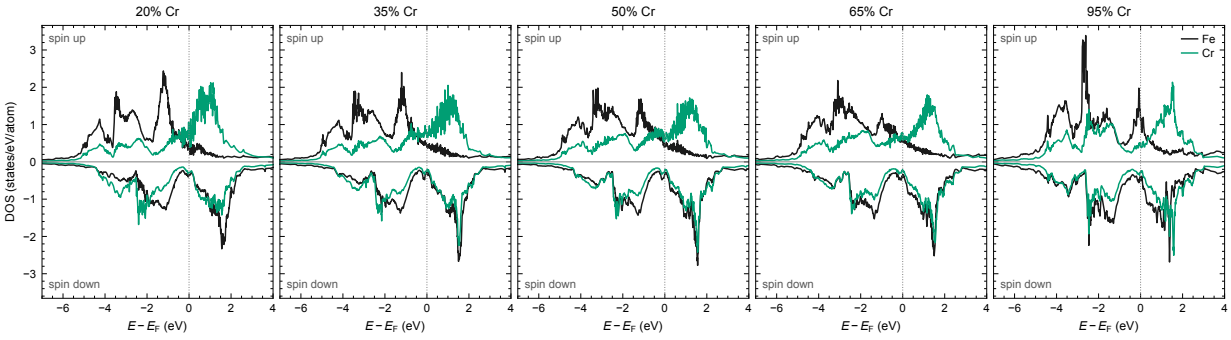


Figure S18: Fe- and Cr-projected densities of states per atom of the relaxed 40-atom SQS (spin up above, spin down below the axis), in the layout of Korzhavyi et al. [7] (their Fig. 4). Validation: the Fermi-level DOS, 0.79–0.98 states/eV/atom at 20–95% Cr, lies 9–21% below the EMTO-CPA ferromagnetic values of Ref. [7] (their Fig. 3d) and follows the same composition trend; the average site moments agree with EMTO-CPA (their Fig. 8) to 0.08 μ_B for Fe and 0.07 μ_B for Cr, and lie within the spread of VASP supercell values of Klaver et al. [38]. The energy window extends to 4 eV above E_F (1.6 times the occupied bands).

S6. Electronic topological transition and the Fe-rich elastic anomaly

The Fe-rich anomaly of the elastic constants (Fig. S8) coincides with a change of the spin-up Fermi-surface topology (Fig. S15; Ref. [7]). It can be described quantitatively as an electronic topological transition (ETT) in the sense of Lifshitz [39,40]. Let E_c be the energy of the critical point, here the triply degenerate majority-spin level at H, and $z = E_F - E_c$. When z changes sign, the electronic grand potential acquires a non-analytic term,

$$\delta\Omega(z) = -c_L\,|z|^{5/2}\,\theta(z), \qquad \delta N(E_F) \propto z^{1/2}\theta(z), \tag{10}$$

so that the singular part of an elastic constant, a second strain derivative of the energy, is

$$\delta C_{ij} = \frac{\partial^2 \delta\Omega}{\partial\varepsilon_i\,\partial\varepsilon_j} \simeq -\tfrac{15}{4}\,c_L\,\frac{\partial z}{\partial\varepsilon_i}\frac{\partial z}{\partial\varepsilon_j}\,z^{1/2}\,\theta(z) + O\!\left(z^{3/2}\right). \tag{11}$$

This is the band term $C_{\mathrm{band}} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2\delta(\varepsilon_k - E_F)$ of the new Fermi-surface sheet. Alloying shifts the critical level approximately linearly, $z \simeq \kappa(x - x_c)$, and substitutional disorder broadens it. Every modulus and the SAW frequency therefore follow

$$Q(x) = a + b\,x + A\left[\mathrm{Re}\,\sqrt{x - x_c + i\Gamma} - \mathrm{Re}\,\sqrt{-x_c + i\Gamma}\right], \tag{12}$$

a smooth background plus a square-root anomaly whose onset x_c and width Γ are common to all quantities; a, b and A differ between them. The sign of A depends on the modulus, because at fixed electron number the shift of E_F also enters. The test of this description is whether one onset, fixed independently by the electronic structure, accounts for every data set (Fig. S19).

Onset from our electronic structure. In ferromagnetic Fe the critical level lies 0.149 eV above E_F at the PBE-relaxed lattice parameter and 0.104 eV above it at the experimental one, so $\partial z/\partial \ln V \simeq 1.07$ eV. In the unfolded bands of the SQS at the experimental lattice parameter (Fig. S17), the weight of this level at H is centred 0.005 eV above E_F at 5% Cr ($z = -0.005$ eV) and 0.009 eV below it at 10% Cr ($z = +0.009$ eV), with disorder spreads of 0.11 and 0.10 eV. A straight line through the three experimental-lattice points gives $\kappa = 1.13$ eV and $x_c = 0.079$, with the spread corresponding to ±0.10 in composition (Fig. S19a). The level reaches E_F already at 5% Cr, so the onset may lie nearer 0.05–0.07 than the linear estimate.

Onset from the elastic constants. A joint fit of Eq. (12) to the EMTO-CPA C_{44}, B, C' and f_{SAW} of Ref. [8] ($0 \le x \le 0.2$), with one shared x_c and Γ, gives $x_c = 0.102$ (0.093–0.108 at $\Delta\chi^2 \le 1$) and $\Gamma = 0.056$. The RMS residuals are 1.2, 2.9 and 0.19 GPa and 1.8 MHz (Fig. S19b–e). Fixing x_c and Γ to the electronic-structure values instead fits all four quantities nearly as well: RMS 1.2, 3.3 and 0.19 GPa and 1.5 MHz, $\Delta\chi^2 = 3.1$ (within two standard deviations of the best fit). The model-independent markers fall in the same window: the knee of the EMTO-CPA C_{44} at $x = 0.059$, the minimum of the EMTO-CPA bulk modulus at 0.109, and the minimum of the measured room-temperature bulk modulus at 0.068 [7]. A secondary minimum of the profile near $x_c \approx 0.02$ ($\Delta\chi^2 = 1.7$), driven by the initial rise of C_{44}, is disfavoured. Our QE C_{44} and f_{SAW} show the same rise and flattening between 0 and 10% Cr (Fig. S19b,d). Only the onset is tested. The amplitudes A are not interpreted, because over this window the background and the singular term are not uniquely separable.

Link to the stiffening on decomposition. The Fe-rich α phase of decomposed Fe–36Cr ($x_\alpha = 0.12$ in MD, 0.14 by CALPHAD at 748 K [34]) lies beyond x_c in every estimate, so decomposition carries the Fe-rich phase across the transition. Splitting the lever-rule gain of Fe–36Cr into the two phases (CALPHAD compositions) gives +7.8% for the electronic EMTO-CPA $f_{\mathrm{SAW}}(x)$, of which α contributes 2.4 and α' 5.4 percentage points, and +5.0% for the classical MD $f_{\mathrm{SAW}}(x)$ (1.4 and 3.6 points; Fig. S19f). With the MD phase compositions the totals are +7.7% and +4.0%, the α terms 2.0 and 1.4 points. The Fe-rich side, where the ETT sits, thus accounts for about 1.0 (0.6) of the 2.8 (3.7) percentage points by which the electronic stiffening exceeds the classical one; the remainder comes from the Cr-rich side. The ETT is therefore a measurable but secondary contributor to the stiffening that TGS detects. To be updated: the critical level at 5 and 10 pct Cr at the PBE-relaxed lattice parameter (unfolded bands running) will add the relaxed-volume estimate; with the same kappa it is about 0.13.

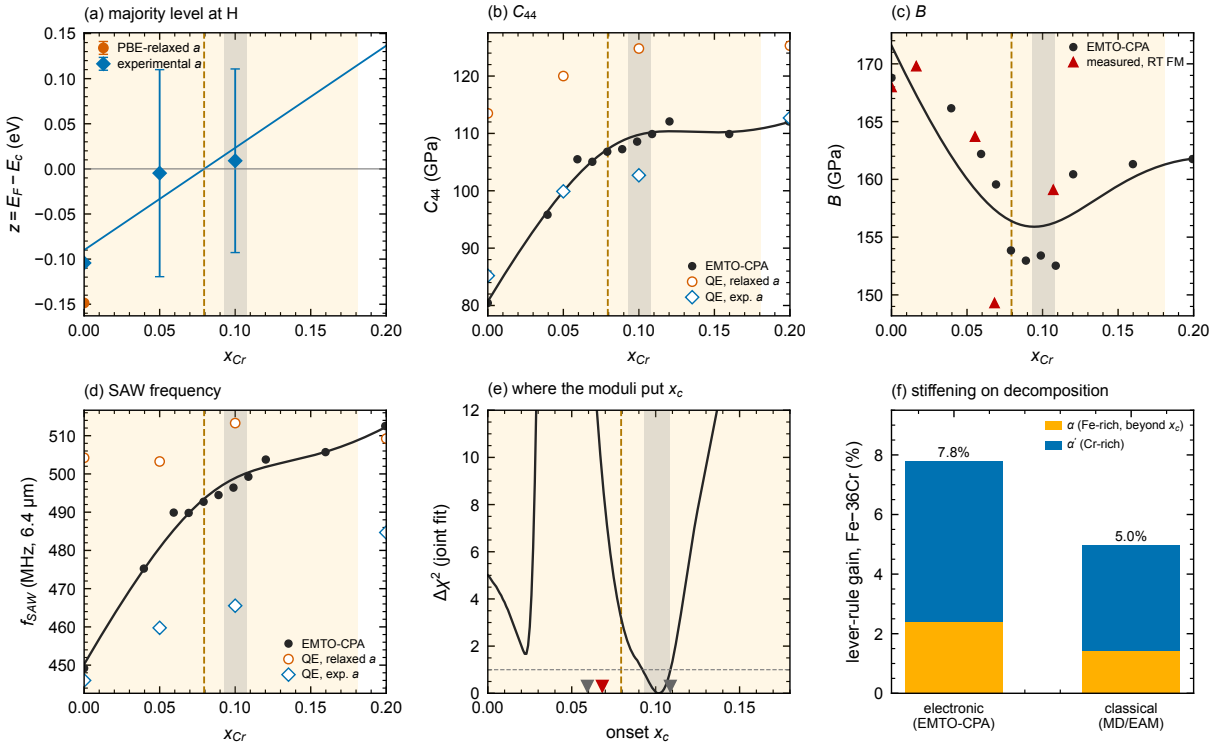


Figure S19: Electronic topological transition on the Fe-rich side. (a) Critical level $z = E_F - E_c$ (majority level at H) from our band structures: pure Fe at both lattice parameters and the unfolded 5 and 10% Cr SQS at the experimental lattice parameter (error bars: disorder spread of the spectral weight). The line through the experimental-lattice points gives $x_c = 0.079$ (dashed; light band: ± spread). (b–d) C_{44}, B and f_{SAW}: EMTO-CPA [8] (black) with the joint fit of Eq. (12), measured bulk moduli (red triangles, [7]) and our QE SQS (open symbols). The grey band is the joint-fit range of x_c. (e) $\Delta\chi^2$ of the joint fit against the onset x_c; triangles mark the model-independent anomaly positions (red: measured). (f) Lever-rule stiffening of Fe–36Cr (CALPHAD phase compositions at 748 K), split into the Fe-rich α and Cr-rich α' terms, for the electronic and classical $f_{SAW}(x)$. Script: dft_qe/ett_analysis.py.

S7. Lattice dynamics

Phonon dispersions of Fe, non-magnetic Cr and antiferromagnetic Cr are shown in Fig. S20, the k-point convergence that removes spurious imaginary modes of non-magnetic Cr in Fig. S21, and the comparison with neutron and nuclear inelastic scattering in Fig. S22. Cr is computed with density-functional perturbation theory; Fe by finite displacements, for the reason given in the captions.

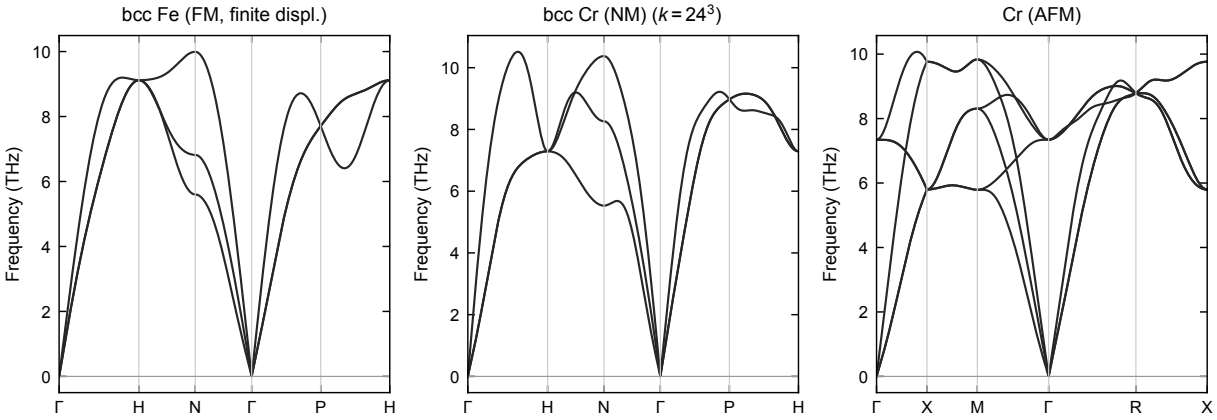


Figure S20: Phonon dispersions of bcc Fe (FM), bcc Cr (NM, DFPT, $k = 24^3$) and commensurate AFM Cr (DFPT, $k = 12^3$, simple-cubic path), at the PBE-relaxed lattice parameters. All frequencies are real. Fe phonons are computed by finite displacements (phonopy; 0.01 Å displacement in a cubic $3 \times 3 \times 3$ conventional supercell of 54 atoms, same PBE-PAW settings, $k = 5^3$), because density-functional perturbation theory (QE ph.x) gave frequencies about 60% too high for ferromagnetic Fe while a frozen phonon with identical settings agreed with neutron scattering to 2% (4.40 vs 4.31 THz for the transverse mode at $\zeta = 1/3$ along Γ–H). A $2 \times 2 \times 5$ supercell was not used: its tetragonal shape made the symmetrised force constants inconsistent and produced a spurious imaginary branch. Cr is computed with DFPT, which agrees with neutron scattering (Fig. S22); its non-magnetic phase needs $k = 24^3$ (Fig. S21).

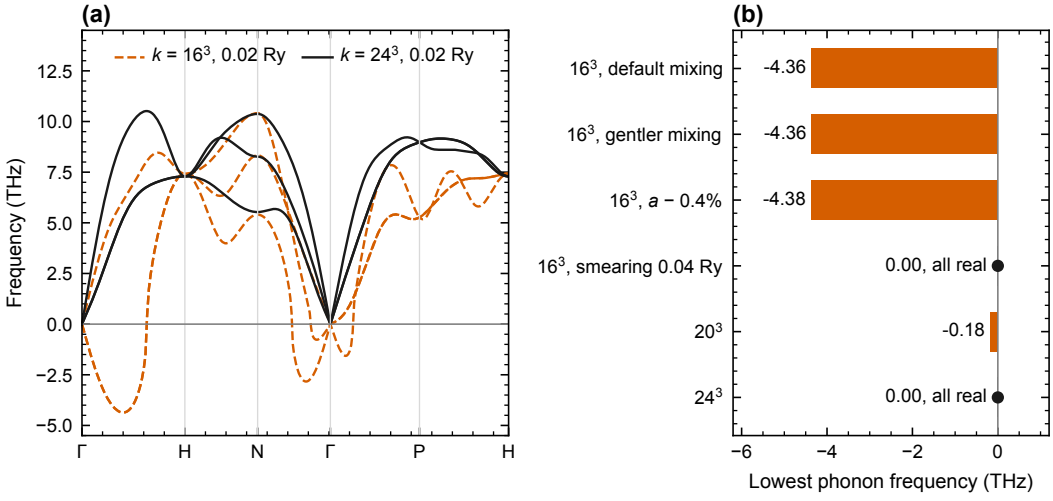


Figure S21: Brillouin-zone sampling of the NM Cr phonons. (a) Dispersion at an electronic k-mesh of 16^3 (imaginary branches plotted below zero) and 24^3 (all real). (b) Lowest frequency along the path for each setting: SCF mixing and a 0.4% change of the lattice parameter have no effect, whereas a denser mesh or wider smearing removes the imaginary modes. They are an artefact of under-sampling the nested Fermi surface of Cr, not a dynamical instability.

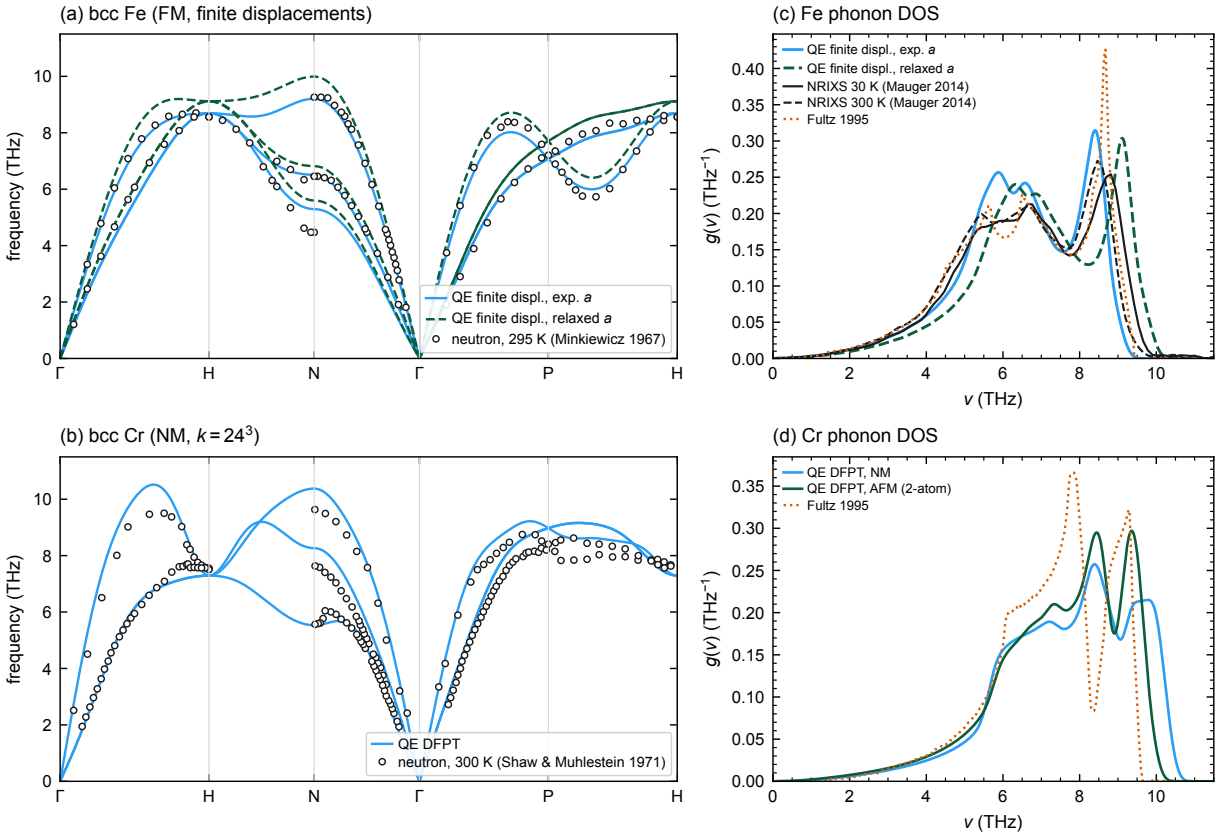


Figure S22: Our phonons against experiment. (a) Fe by finite displacements at the experimental (solid) and PBE-relaxed (dashed) lattice parameter, with the room-temperature neutron dispersion of Minkiewicz et al. [41]; (b) non-magnetic Cr (DFPT, $k = 24^3$) with the 300 K neutron data of Shaw and Muhlestein [42]; (c) Fe phonon DOS against nuclear inelastic scattering at 30 and 300 K (Mauger et al. [43]) and the force-constant DOS of Fultz et al. [32]; (d) Cr phonon DOS (NM and AFM) against Fultz et al. Agreement (nearest-branch RMS deviation from the neutron data; mean frequency against the reference DOS): Fe 0.23 THz and −0.1% at the experimental lattice parameter, 0.46 THz and +7.8% at the relaxed one; Cr 0.41 THz (NM) and 0.35 THz (AFM), +6% and +3%. Fe phonons are computed by finite displacements (phonopy; 0.01 Å displacement in a cubic $3 \times 3 \times 3$ conventional supercell of 54 atoms, same PBE-PAW settings, $k = 5^3$), because density-functional perturbation theory (QE ph.x) gave frequencies about 60% too high for ferromagnetic Fe while a frozen phonon with identical settings agreed with neutron scattering to 2% (4.40 vs 4.31 THz for the transverse mode at $\zeta = 1/3$ along Γ–H). A $2 \times 2 \times 5$ supercell was not used: its tetragonal shape made the symmetrised force constants inconsistent and produced a spurious imaginary branch.

S8. Validation against literature and experiment

Every quantity shown in Sections S4–S7 was replotted on top of published data it should reproduce (Figs. S23–S28). Literature curves were digitized from the published figures (raster images; the uncertainty of each digitization is given with the data in the repository, and each was checked by overlaying it on the original figure). Deviations are summarised in Table S1; the full list, including every composition, is in validation_summary.csv.

Table S1: Validation of the first-principles calculations. RMS: root-mean-square deviation; for dispersions, the deviation of each experimental point from the nearest calculated branch. "Relaxed": PBE-relaxed lattice parameter; "exp. a": room-temperature experimental lattice parameter.

Quantity	Reference	Agreement
Bulk modulus	measured, compiled in Ref. [7]	relaxed: +28% (QE Fe), +31% (QE, 95% Cr), +32% (VASP Fe); exp. a: +1% (QE Fe), −7% (QE AFM Cr), +7% (QE, 95% Cr)
Site moments	EMTO-CPA FM [7]; VASP supercells [38]	mean $\lvert\Delta m\rvert$: Fe 0.08, Cr 0.07 μ_B (relaxed); Fe 0.18, Cr 0.03 μ_B (exp. a); within the supercell spread
$N(E_F)$	EMTO-CPA FM [7]	9–21% lower, same composition trend
Fe bands, Γ–H	EMTO-CPA, pure Fe [7]	RMS 0.22/0.17 eV (spin up/down, relaxed); 0.18/0.14 eV (exp. a)
Cr phonons	neutron, 300 K [42]	RMS 0.35–0.41 THz, bias −0.05 to +0.17 THz (NM and AFM, both lattice parameters)
Cr phonon DOS	force-constant DOS [32]	mean frequency +6% (NM), +3% (AFM); exp. a: −0.4%, +1.0%
Fe phonons	neutron, 295 K [41]	finite displacements: RMS 0.23 THz (exp. a), 0.46 THz (relaxed); DFPT: 1.1–1.4 THz (not used)
Fe phonon DOS	NRIXS, 30 and 300 K [43]	mean frequency −0.1% (exp. a), +7.8% (relaxed); DFPT +19 to +27%

The electronic structure and magnetism agree well with the literature, and the Cr lattice dynamics agree with neutron scattering. The bulk modulus is the important exception. At the PBE-relaxed volume both DFT codes overestimate the measured bulk modulus by about 30%, the usual consequence of the too-small PBE volume of Fe–Cr; at the room-temperature experimental lattice parameter the QE value at 95% Cr is within 7% of experiment. The same setting reproduces the minimum of the bulk modulus and the plateau of f_{SAW} near 40% Cr reported with EMTO-CPA at experimental lattice parameters [8], which argues for basing the QE elastic constants on the experimental-lattice-parameter series. For Fe, density-functional perturbation theory is too stiff at both lattice parameters (Figs. S29 and S30), whereas finite displacements with the same pseudopotential, cutoff and k-point density agree with neutron and NRIXS data; the Fe phonons are therefore taken from finite displacements (Fig. S20). To be updated: the pure-metal electronic DOS against Fig. 4 of Korzhavyi et al. will be added.

Thermal transport (Section S1).. The estimates of Section S1 depend only on E, ρ and the molar mass and were evaluated for every tensor of this work (Fig. S31). The elastic Debye temperature of Fe is 450 K (QE, experimental lattice parameter), 453 K (EMTO-CPA) and 472 K (MD), against 470 K from calorimetry [35]; for Cr, 585–605 K against 630 K. Debye temperatures from the second moment of the phonon DOS, a definition that runs lower, agree with experiment on the same definition: 415 K for our Fe against 417–431 K

from NRIXS, and 466 K for our AFM Cr against 460 K from the neutron-based DOS of Fultz et al. The minimum thermal conductivity of Eq. (2), κ_{min} = 1.15–1.6 W/(m K), gives $D_{min} = \kappa_{min}/\rho c_p$ = 0.32–0.48 × 10^{-6} m^2/s, an order of magnitude below the thermal diffusivity measured by TGS in CF8 (3.5 × 10^{-6} m^2/s, Fig. S7d). This is expected, since κ_{min} is a lower bound on the lattice contribution and electrons carry most of the heat in a metal. The relative changes do not depend on the prefactor. On full decomposition of MD Fe–36Cr, E rises by 9.5% and the S1 scaling $D \propto E^{1/2}$ gives $\Delta D/D$ = +4.6%, close to the +4.0% measured after aging at 290 °C. The larger increases measured after aging at 330–400 °C (+20 to +29%) exceed what lattice stiffening alone can produce; they point to the electronic contribution, which the S1 estimate does not include. Panels (e) and (f) of Fig. S7 now follow Eq. (2) as printed: the earlier version of the working sheet had evaluated κ with Planck's constant (×10^{11}) in place of k_B and with m = 2, which overstated κ and D by a factor of 7.6 without changing their trend.

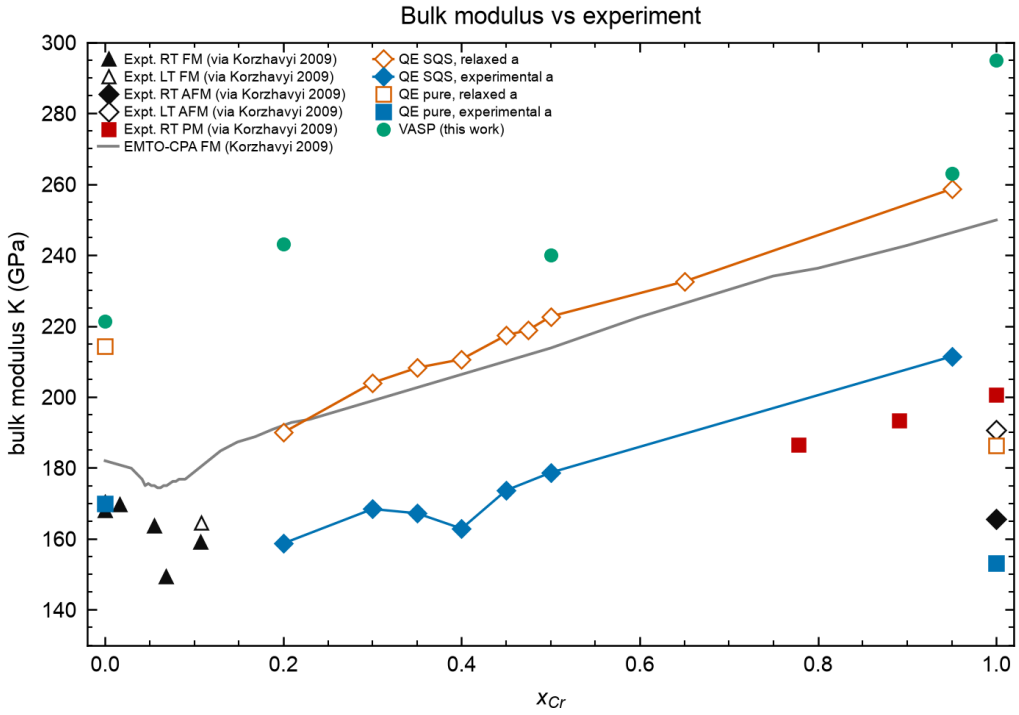


Figure S23: Bulk modulus against experiment. Markers: measured values compiled by Korzhavyi et al. [7] (RT and LT; FM, AFM and paramagnetic); grey line: their EMTO-CPA FM calculation. QE SQS (diamonds) and pure elements (squares; 2-atom cells, $k = 16^3$, Fe FM and Cr AFM) at the PBE-relaxed (open) and the room-temperature experimental (filled) lattice parameter; VASP (circles). At the relaxed volume both codes overestimate B: QE by +28% (Fe) and +31% (95% Cr), VASP by +32% (Fe). At the experimental lattice parameter QE gives B = 170 GPa for Fe (+1% against 168 GPa), −7% for AFM Cr and +7% at 95% Cr. The experimental lattice parameter was therefore adopted for the additional QE series of Section S4; PBE underestimates the Fe–Cr volume by about 4%, and the bulk modulus is the property most sensitive to it.

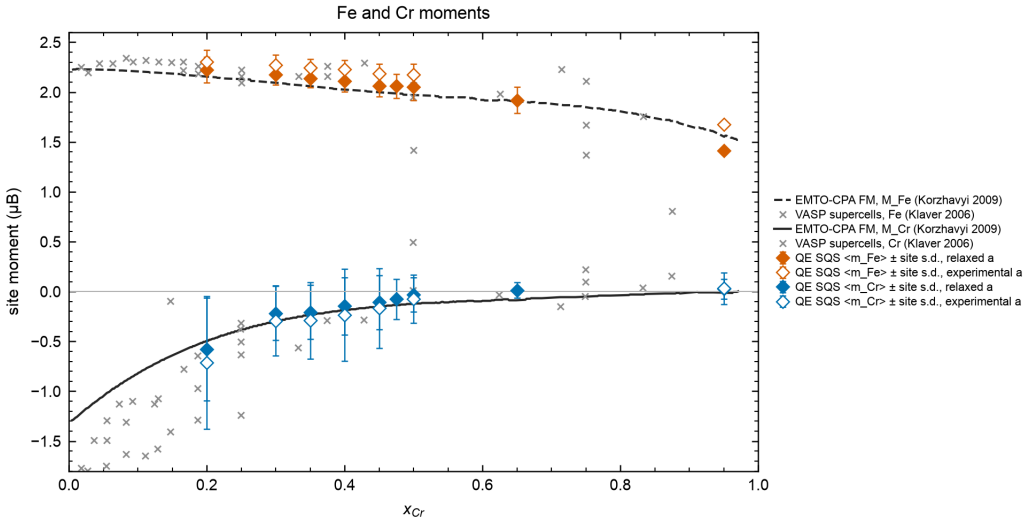


Figure S24: Average Fe and Cr site moments of the QE SQS (error bars: standard deviation over sites) at the relaxed (filled) and experimental (open) lattice parameter, against EMTO-CPA FM (lines; Korzhavyi et al. [7], Fig. 8, a_0 = 2.875 Å) and VASP supercells (crosses; Klaver et al. [38]). The collapse of the Cr moment between 20 and 50% Cr is reproduced.

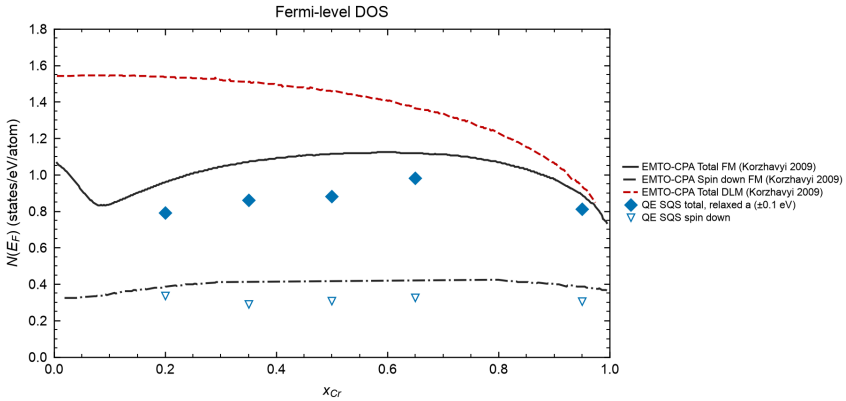


Figure S25: Fermi-level density of states per atom of the relaxed QE SQS (total and spin down, averaged over ±0.1 eV) against EMTO-CPA for the FM and paramagnetic (DLM) states (Korzhavyi et al. [7], Fig. 3d). The QE values are 9–21% lower and follow the FM composition trend.

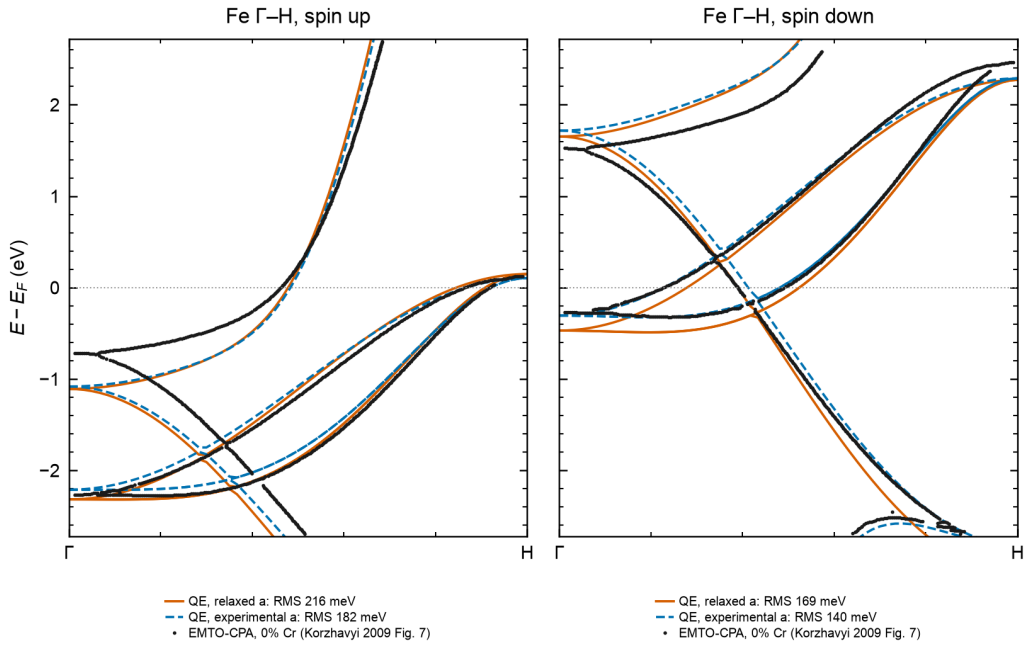


Figure S26: Band structure of FM bcc Fe along Γ–H, majority (left) and minority (right), QE at the relaxed and experimental lattice parameters, against the EMTO-CPA spectrum of pure Fe (points; Korzhavyi et al. [7], Fig. 7a,d; lattice parameter not stated). RMS deviations 0.14–0.22 eV; the Fermi-level crossings and the majority H level 0.15 eV above E_F agree.

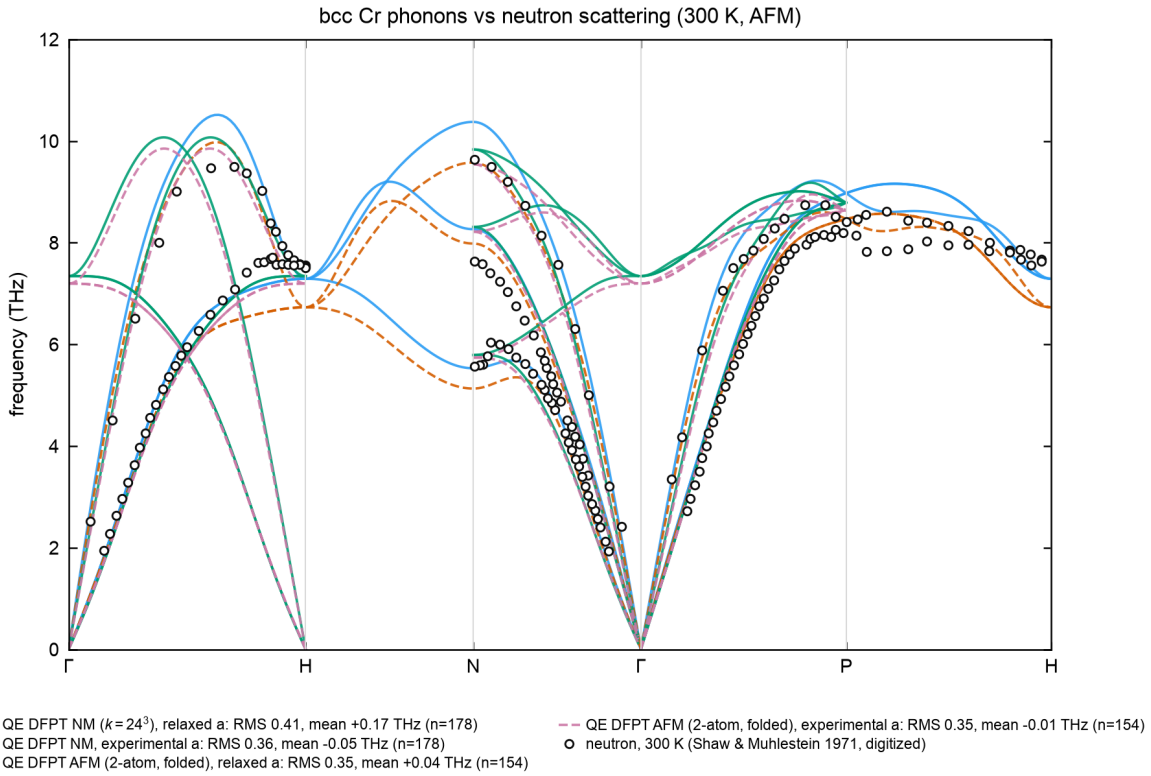


Figure S27: Phonon dispersion of Cr against neutron scattering at 300 K (Shaw and Muhlestein [42], digitized; Cr is AFM at 300 K): DFPT for NM Cr ($k = 24^3$) and commensurate AFM Cr (2-atom simple-cubic cell, branches folded onto the bcc path where the lines coincide), each at the relaxed and the experimental lattice parameter. RMS deviations 0.35–0.41 THz.

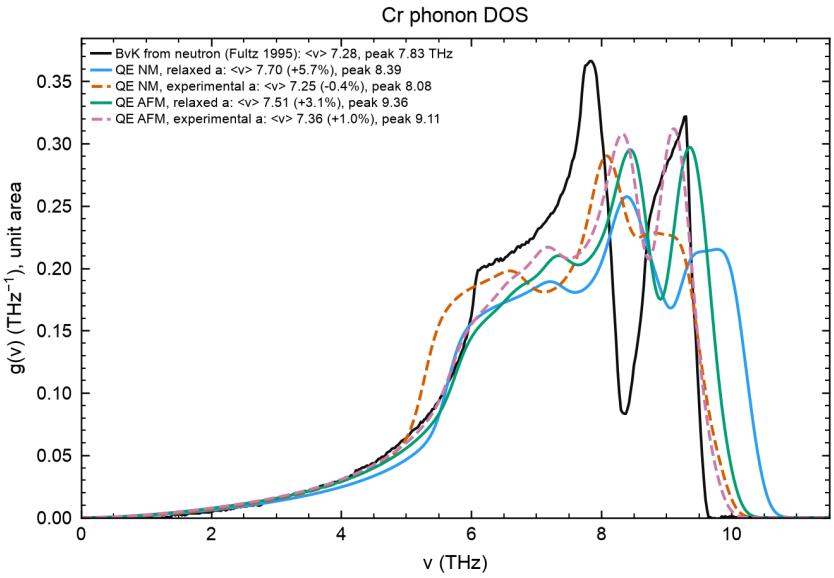


Figure S28: Phonon DOS of Cr (DFPT, NM and AFM, unit area) against the Born–von Kármán DOS of Fultz et al. [32], which is calculated from force constants fitted to neutron dispersions (not a measured spectrum). Mean frequencies are 6% (NM) and 3% (AFM) higher.

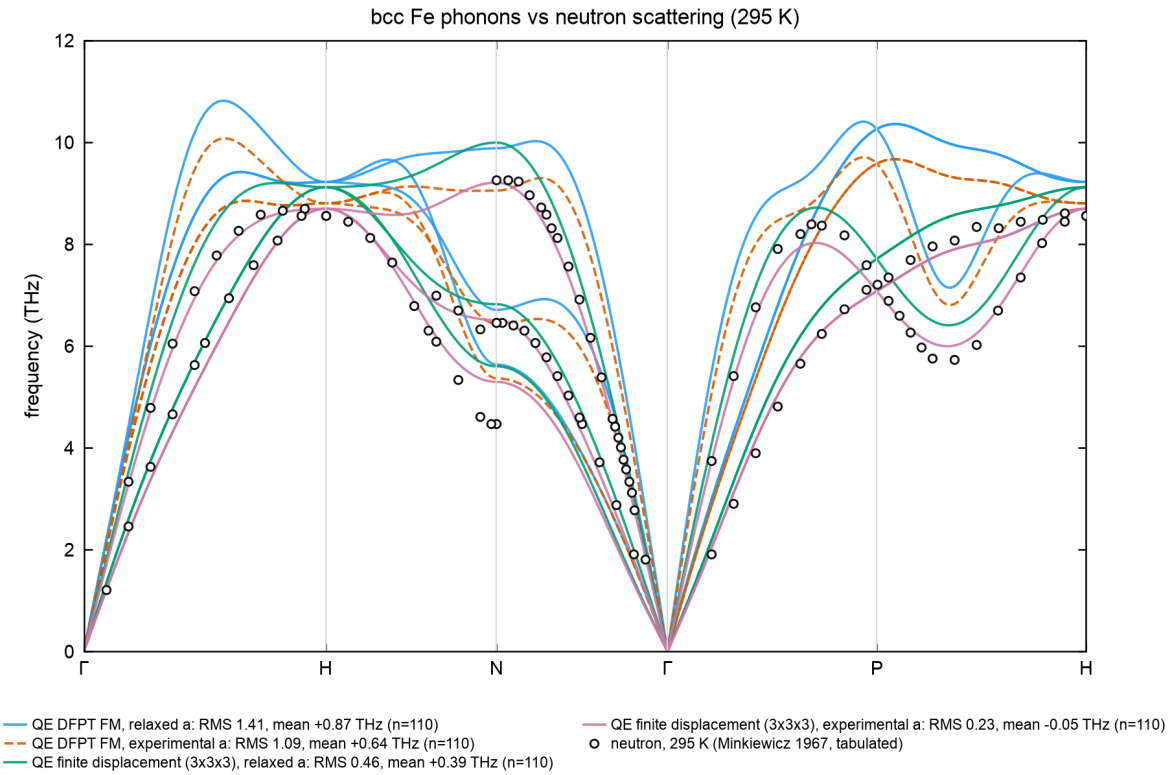


Figure S29: Phonon dispersion of FM Fe against neutron scattering at 295 K (Minkiewicz et al. [41], tabulated): DFPT and finite displacements, each at the relaxed and the experimental lattice parameter. DFPT is too stiff (RMS 1.4 and 1.1 THz); finite displacements agree (RMS 0.46 and 0.23 THz). Fe phonons are computed by finite displacements (phonopy; 0.01 Å displacement in a cubic $3 \times 3 \times 3$ conventional supercell of 54 atoms, same PBE-PAW settings, $k = 5^3$), because density-functional perturbation theory (QE ph.x) gave frequencies about 60% too high for ferromagnetic Fe while a frozen phonon with identical settings agreed with neutron scattering to 2% (4.40 vs 4.31 THz for the transverse mode at $\zeta = 1/3$ along Γ–H). A $2 \times 2 \times 5$ supercell was not used: its tetragonal shape made the symmetrised force constants inconsistent and produced a spurious imaginary branch.

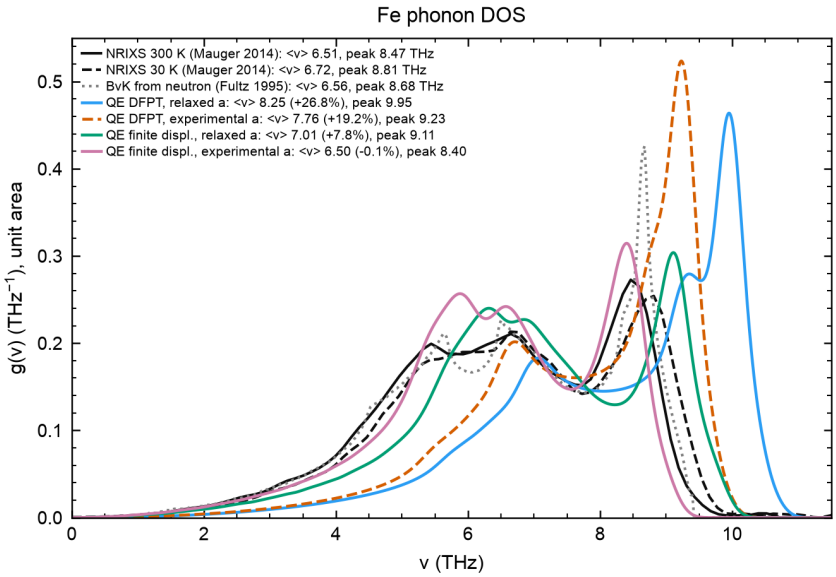


Figure S30: Phonon DOS of FM Fe (unit area) against NRIXS at 30 and 300 K (Mauger et al. [43]) and the force-constant DOS of Fultz et al. [32]. Mean frequencies relative to the 300 K NRIXS DOS: DFPT +27% (relaxed) and +19% (experimental lattice parameter); finite displacements +7.8% and −0.1%. The finite-displacement result at the experimental lattice parameter is used for Fe.

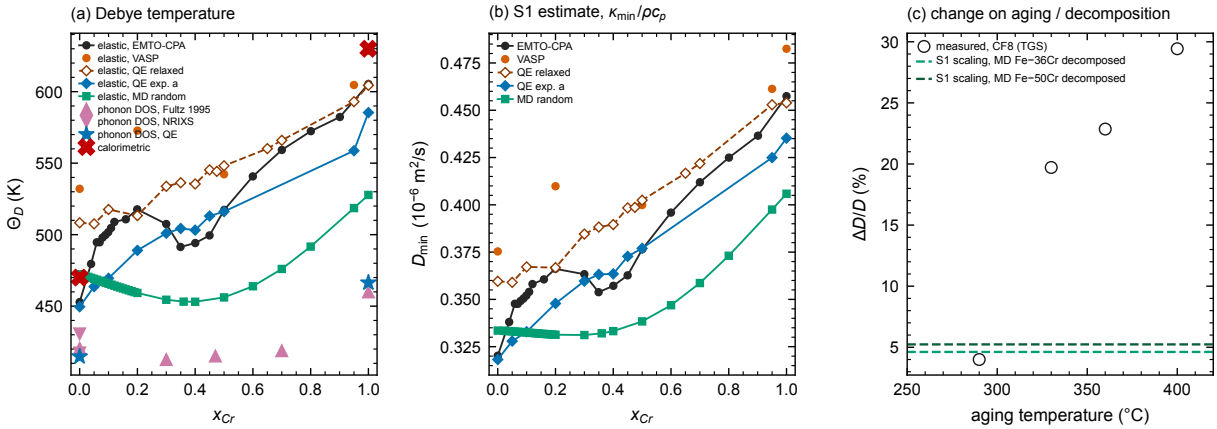


Figure S31: Validation of the thermal-transport estimates of Section S1. (a) Debye temperature: elastic (Anderson) from every tensor of this work and EMTO-CPA; from the second moment of the phonon DOS (Fultz et al. [32], NRIXS [43], ours; this definition runs about 10% below the elastic one); calorimetric values of Fe and Cr [35]. (b) $D_{\min} = \kappa_{\min}/\rho c_p$ from Eq. (2) of Section S1 for each tensor set; it is a lower bound on the lattice contribution. (c) Relative change of the thermal diffusivity: measured by TGS in CF8 after aging (circles, Fig. S7d) and predicted by the S1 scaling for full decomposition of MD Fe–36Cr and Fe–50Cr (dashed). Script: dft_qe/thermal_check.py.

S9. Proposed mechanism figures

The two figures below collect the mechanism behind the anomalous elasticity and the stiffening on decomposition (Fig. S32), and its consequence for embrittlement (Fig. S34). They are placed here for now; we may move them to the main text.

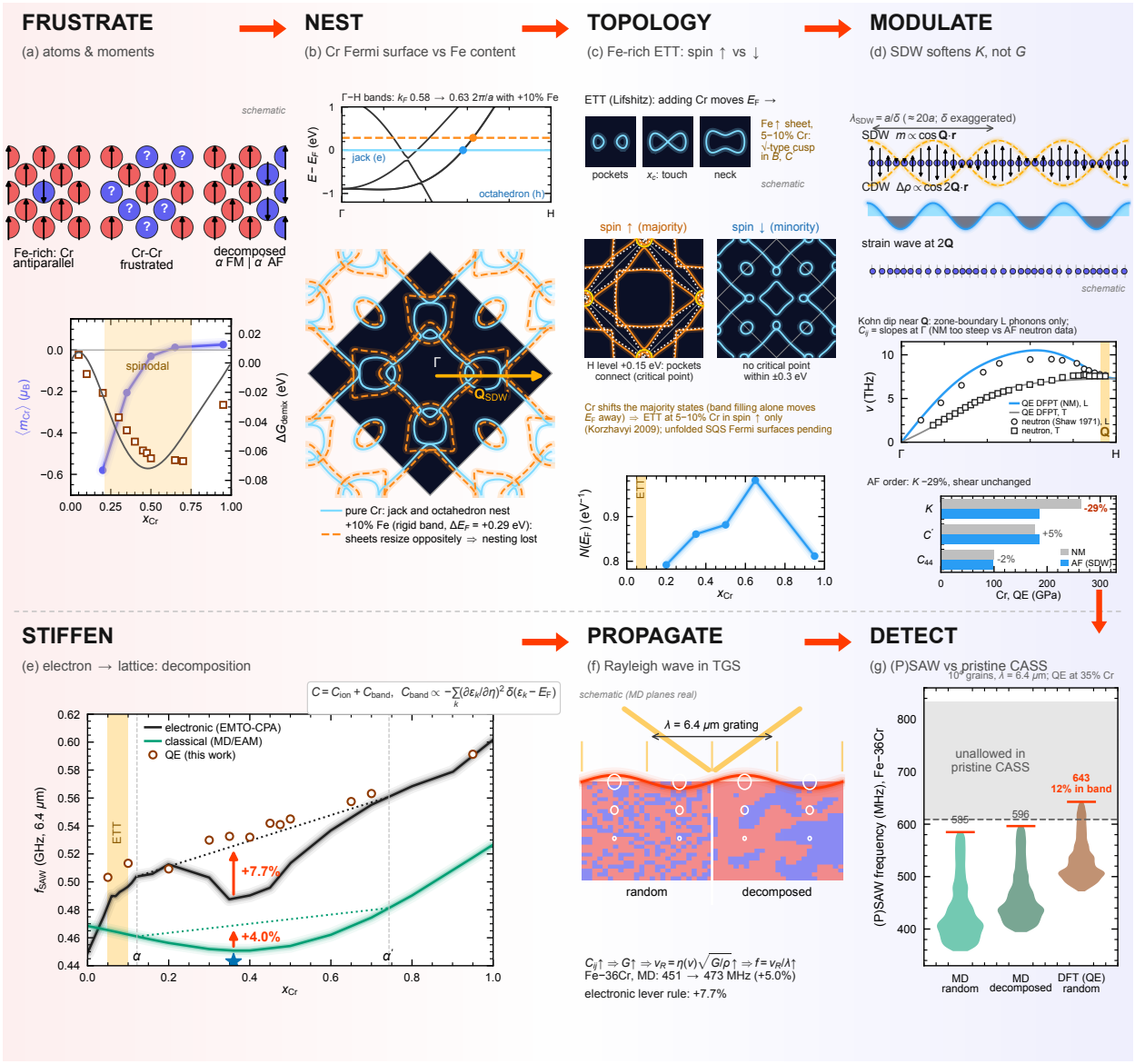


Figure S32: Mechanism of the anomalous elasticity of bcc Fe–Cr and of the stiffening on spinodal decomposition. Panel descriptions are on the following page.

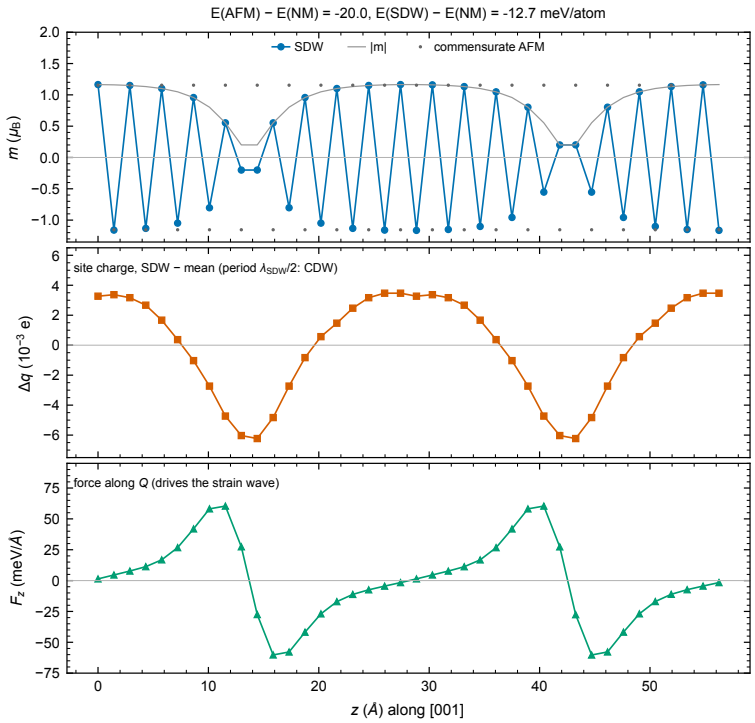


Figure S33: Spin- and charge-density wave of bcc Cr computed in a $1 \times 1 \times 20$ supercell of the conventional cell (40 atoms along [001], room-temperature experimental $a = 2.8846$ Å, $k = 16 \times 16 \times 1$, other settings as the SQS runs). The supercell sets the incommensurability $\delta = 1/20$, close to the measured 0.048 [44]. Top: site moments of the SDW, which has nodes every $\lambda_{SDW}/2 = 29$ Å, with the commensurate AFM moments (dots) for comparison. Middle: site charges relative to their mean. They oscillate with period $\lambda_{SDW}/2$, the charge-density wave that accompanies the SDW, with a peak-to-peak amplitude of about 0.01 e. Bottom: forces along Q at the unrelaxed positions, also with period $\lambda_{SDW}/2$; they would drive the strain wave seen by X-ray diffraction [44]. In PBE the SDW is 12.7 meV/atom below the non-magnetic state but 7.3 meV/atom above the commensurate AFM, and its moments (up to 1.16 μ_B) exceed the measured 0.62 μ_B. Both are known shortcomings of semilocal DFT for Cr [45,46], so the figure shows the form of the coupled spin, charge and strain waves, not their measured amplitudes. Script: dft_ge/sdw_cr.py.

Figure S32 (continued). (a) Magnetic frustration: isolated Cr is antiparallel to Fe; Cr–Cr neighbours are frustrated and the Cr moments collapse (Quantum ESPRESSO SQS: $\langle m_{Cr} \rangle = -0.58, -0.21, -0.03\,\mu_B$ at 20, 35, 50% Cr), giving $\Delta G_{demix} < 0$ and the spinodal window (QE squares, PWSCF–PAW model line). (b) Top: non-magnetic Cr bands along Γ–H (QE); the d band that bounds the jack and the octahedron crosses E_F at $k_F = 0.58\,(2\pi/a)$, moving to 0.63 with a rigid-band 10% Fe. Bottom: Fermi surface of non-magnetic Cr (QE, $k_z = 0$): the electron jack at Γ nests onto the hole octahedron at H with $\mathbf{Q}_{SDW}$; adding Fe (rigid band, $N(E_F)$ of Ref. [47]) resizes the two sheets in opposite senses and destroys the nesting (Fe makes the SDW of Cr commensurate at about 2.3 at.% and lowers T_N by 15–20 K per at.% [44]). (c) Fe-rich side, by spin: Fermi-surface sections of ferromagnetic Fe (QE). In the majority channel a band level at H lies 0.15 eV above E_F, where the H pockets connect (dotted contour, circles): this is the critical point of the electronic topological transition that Korzhavyi et al. find in the spin-up Fermi surface at 5–10% Cr [7] (with the bulk-modulus anomaly of Ref. [48]). The minority channel has no critical point within ±0.3 eV. Rigid filling would move E_F down, away from the critical level, so the transition is driven by the Cr-induced shift of the majority states rather than by band filling (inset: schematic Lifshitz sequence). (d) In pure Cr the SDW, with its charge-density and strain waves at $2\mathbf{Q}$ [44] (schematic), leaves two lattice signatures, and neither is the stiffening. The nesting softens the longitudinal phonons near $\mathbf{Q}$ (Kohn dip; DFPT for non-magnetic Cr at $k = 24^3$, and neutron scattering at 300 K [42]), but only near the zone boundary; the elastic constants are the slopes at Γ. Antiferromagnetic order softens the volume channel, not shear (QE, pure Cr): C_{12} falls from 146 to 63 GPa and K from 264 to 186 GPa (−29%), while C' (177 → 185 GPa) and C_{44} (100 → 98 GPa) barely change. Fe brings T_N below room temperature at about 1 at.% Fe and suppresses antiferromagnetism near 18 at.% Fe [34], so the Cr-rich α' of decomposed Fe–36Cr ($x \approx 0.74$ in MD, 0.88 by CALPHAD at 748 K) is paramagnetic when TGS measures it. The stiffening on decomposition is a shear effect (e; Fig. S34). (e) Decomposition places the Fe-rich α ($x = 0.12$) and Cr-rich α' ($x = 0.74$) phases of MD Fe–36Cr on stiffer parts of a convex $f_{SAW}(x)$: the lever rule gives +4.0% with the MD (classical) curve, which accounts for most of the +5.0% obtained for the fully decomposed MD structure, and +7.7% with the EMTO-CPA (electronic) curve whose random-alloy minimum lies near Fe–36Cr. With the CALPHAD phase compositions at 748 K (α 0.14, α' 0.88 [34]) these become +5.0% (MD, matching the +5.0% of the decomposed MD structure) and +7.8% (EMTO-CPA). The relaxed-volume QE curve, which has no minimum, gives no lever-rule gain (−0.3%). (f) Schematic of the TGS measurement over the MD microstructures (random and decomposed, real (001) planes). (g) (P)SAW frequencies of 10^4 grains at $\lambda = 6.4\,\mu$m: Fe–36Cr from MD (random, decomposed) and Fe–35Cr from QE; the QE tensor puts 11.7% of grains above the 609 MHz band edge. To be updated: (b) uses a rigid-band shift based on the Moruzzi Fermi-level density of states; it will be replaced by our full nesting function. (c) will add the unfolded SQS Fermi surfaces at 5 and 10 at.-pct Cr. (d) will be redrawn from the computed spin- and charge-density-wave profile of our Cr supercell, which is now in the SDW figure that follows.

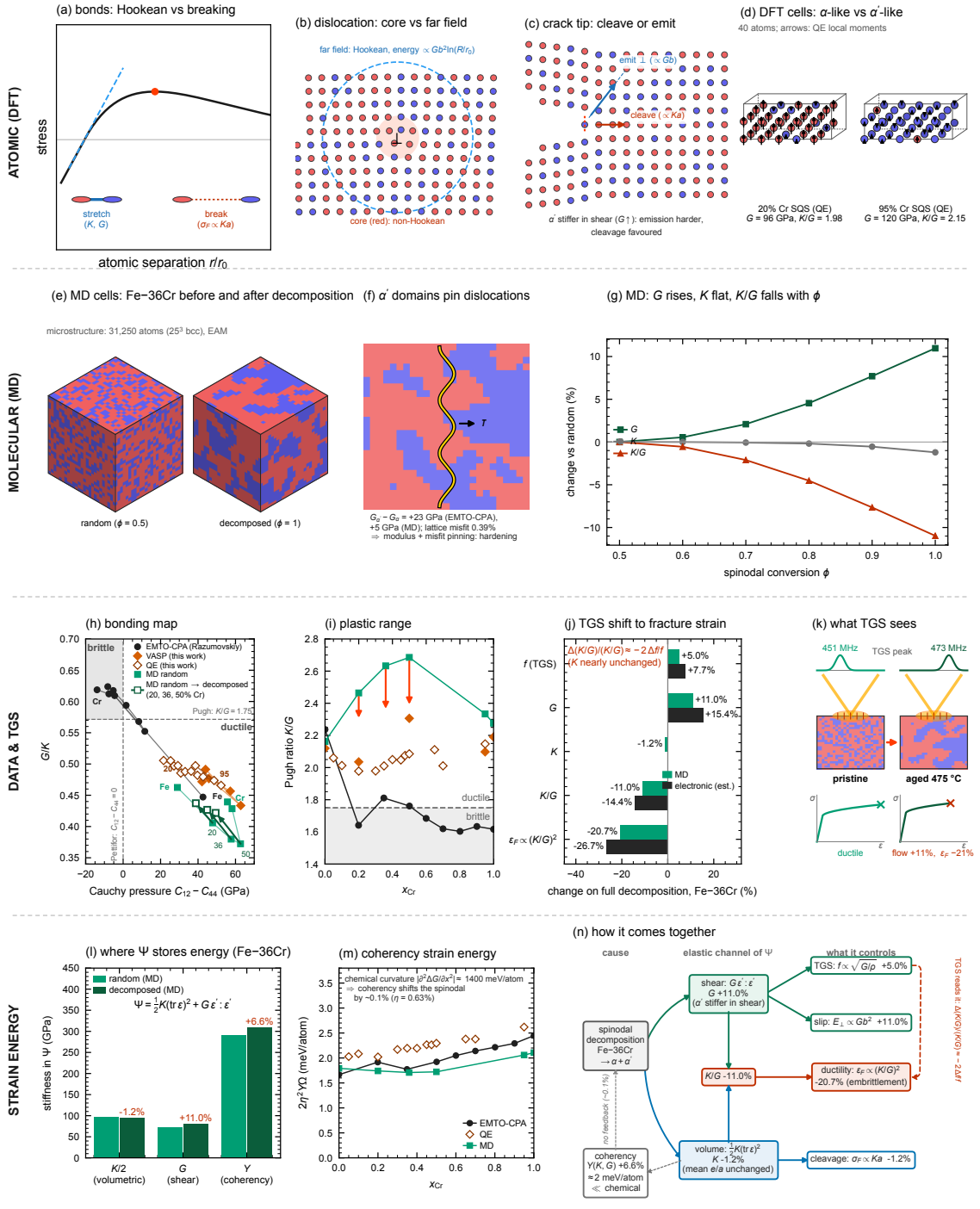


Figure S34: From elastic stiffening to embrittlement, after Pugh [4], at the atomic (DFT) and molecular (MD) scales. Panel descriptions are on the following page.

Figure S34 (continued). Atomic row (schematic except (d)). (a) A bond stretches in the Hookean regime (slope $\propto K$, and G in shear) and breaks at the non-Hookean peak of the stress–separation curve, so fracture strength scales with Ka. (b) An edge dislocation in a random Fe–Cr lattice: its energy, $\propto Gb^2 \ln(R/r_0)$, is stored mostly in the Hookean far field, so slip resistance scales with Gb. (c) At a crack tip the alloy either cleaves ($\propto Ka$) or emits a dislocation ($\propto Gb$); a phase stiffer in shear makes emission harder and favours cleavage. (d) The relaxed 40-atom QE cells of an α-like (20% Cr) and an α'-like (95% Cr) composition with their local moments: G rises from 96 to 120 GPa, while their individual K/G (1.98, 2.15) stays above Pugh's boundary. Molecular row. (e) The 25^3 MD cells of Fe–36Cr, random ($\phi = 0.5$) and fully decomposed ($\phi = 1$). (f) A dislocation bowing between α' domains on the decomposed MD plane: the α/α' shear-modulus misfit is +23 GPa with the EMTO-CPA moduli but about +5 GPa with MD, and the lattice misfit is 0.39%. (g) MD Fe–36Cr: with increasing conversion ϕ, G rises by 11%, K stays within 1% and K/G falls by 11%. Data and TGS. (h) Bonding map: G/K against the Cauchy pressure $C_{12} - C_{44}$ [49], each series joined in composition order (labels: at.% Cr). Dashed lines: Pettifor's ($C_{12} - C_{44} = 0$) and Pugh's ($K/G = 1.75$) criteria; only the quadrant brittle by both is shaded. EMTO-CPA moves from ductile Fe into that quadrant on the Cr-rich side, whereas every tensor of this work (VASP, QE, MD) stays ductile; the MD paths from random to fully decomposed (20, 36, 50% Cr; open squares) all point towards brittle without crossing either line. (i) Pugh ratio versus composition, with the empirical ductile/brittle boundary 1.75. (j) Changes on full decomposition of Fe–36Cr: $\Delta(K/G)/(K/G) \approx -2\,\Delta f/f$; with $\varepsilon_F \propto (K/G)^2$ the fracture strain falls by about 21% (MD) or 27% (electronic lever-rule estimate). (k) What TGS sees: the peak moves from 451 to 474 MHz while the flow stress rises ($\propto Gb$) and the plastic range shrinks (schematic stress–strain curves). Strain energy (bottom row). The elastic energy density $\Psi(K,G) = \frac{1}{2}K(\mathrm{tr}\,\varepsilon)^2 + G\,\varepsilon' : \varepsilon'$ links every step. (l) On decomposition of Fe–36Cr (MD) the volumetric stiffness changes by −1.2%, the shear stiffness by +11.0% and Cahn's coherency modulus $Y = 18KG/(3K + 4G)$ by +6.6%. (m) The coherency strain energy $2\eta^2 Y\Omega$ is about 2 meV/atom for all tensors ($\eta = 0.63\%$), three orders of magnitude below the chemical curvature, so elasticity barely shifts the Fe–Cr spinodal. (n) Mechanism, read left to right: decomposition changes the shear channel of Ψ (G +11.0%; α' is stiffer in shear) but hardly the volume channel (K −1.2%), so K/G falls by 11.0%. The shear channel sets the TGS frequency ($f = \eta_R(\nu)\sqrt{G/\rho}/\lambda$, +5.0%) and the slip resistance ($\propto Gb^2$, +11.0%); the volume channel sets cleavage ($\propto Ka$, −1.2%); K/G sets ductility ($\varepsilon_F \propto (K/G)^2$, −20.7%). Because K is nearly fixed, the TGS shift reads embrittlement directly, $\Delta(K/G)/(K/G) \approx -2\,\Delta f/f$ (dashed). Coherency (Y +6.6%; about 2 meV/atom against about 1400 meV/atom of chemical curvature) feeds back on the decomposition by only about 0.1%. Pugh's relations are empirical correlations for pure metals; the numbers indicate direction and scale, not a prediction for CASS. To be updated: the QE phase tensors at the experimental lattice parameter and QE Fe-80Cr will be added; the MD numbers are final.